

TrackMaster Pro

User Manual · Version 1.5.0

Track planning for model railway layouts in all common gauges – with height planning, helix and timber list, modules, levels and benchwork, scenery, room planning, wiring and signals, plan check, costs and 3D view. For Windows, Android and the browser – with a step-by-step introduction for everyone planning for the first time.

TrackMaster Pro 1.5.0 · Basic and Pro

As of November 2026

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1 Welcome

TrackMaster Pro is a planning program for model railway layouts. You set up the baseboard, lay track from the catalogues of the common track systems, plan gradients and helixes, design the scenery, wire the layout and look at the result in 3D – before you buy the first piece of rail.

What TrackMaster Pro can do

- **Track plans in all common gauges:** Z, N, TT, H0, H0m, H0e, 0, gauge 1 and G, with many track systems and about 900 track pieces – straights, curves, turnouts, crossings, double slips, buffer stops and flexible track.
- **Baseboards in many shapes:** rectangle, L, U, T, Z, E and G shape, or a ring with an open centre – each leg with its own dimensions – or a layout made of **modules** that can be split, attached, moved and rotated.
- **Levels:** up to 10 levels with their own base height, each one can be shown or hidden; the terrain follows.
- **Benchwork:** a timber frame with legs including material and cutting lists; **rib construction** and **roadbeds** for open-grid benchwork.
- **Free-form flexible track:** start at a track end and drag; the track bends along and snaps onto the target.
- **Height planning:** gradients over whole track sequences, round or oval helixes, single or double track.
- **Timber list for the helix:** all cut pieces with edge allowance and saw allowance, supports, fixings and sheet requirement.
- **Scenery:** hills and mountains, water, buildings, trees and tunnels.
- **Wiring and digital:** feeders, isolating joints, power sections, reverse loop detection, light signals with prototypical aspects, detectors, decoder addresses, routes and a wiring plan.
- **Room:** the room with walls, doors, windows, radiators and furniture around the layout – the check reports aisles that are too narrow and blocked doors.
- **Background image:** put a floor plan, sketch or old track plan under the plan and scale it exactly with a dimension line.
- **Help with laying track:** parallel track at a fixed spacing, height profile as a chart, locking, grouping and aligning tracks, usable length of sidings.
- **Plan check:** track spacing, clearance heights, baseboard edge, gradients, scenery, room, usable lengths, open track ends and checks for long vehicles.
- **3D view:** the layout from every side and from below, with benchwork and embankments, a cab ride, a running train and light according to the time of day.
- **Your way of working:** view rotatable to any angle, five designs, seven fonts in four sizes, German or English interface.
- **Results** PRO saving large plans (more than 50 tracks), parts list with prices and your own stock, helix timber list, rib construction, roadbeds, printing and exports (SVG, PNG, DXF).

Basic and Pro

TrackMaster Pro comes in two editions. Without a licence, the **Basic edition** runs. It contains **all planning features** – so you can design, check and view any layout in 3D in full. A licence file turns it into the **Pro edition**, which unlocks the results of your planning.

Basic (without licence)	Pro (with licence) – in addition
Track planning in all gauges with all track systems, flexible track	Saving plans of any size (Basic: up to 50 tracks)
Baseboard shapes, modules, leg dimensions, levels, benchwork with material list	Parts list of the tracks with accessories, also as a CSV file
Height planning: gradients, round and oval helixes, single and double track	Rib construction of the layout and roadbed from selection, with SVG cutting templates
Terrain, water, buildings, vegetation, tunnels	Helix timber list with all cut pieces, also as SVG
Wiring: feeders, isolating joints, signals, decoders, routes, wiring plan	Printing the track plan, also as a 1:1 building template
3D view with cab ride, train and time of day, plan and wiring check	Export as SVG at 1:1, wiring plan as text, export for control software
Room with doors, windows and furniture, background image	Parts list with prices, stock, “to buy” and price list import
Parallel track, height profile, lock, group, align, detectors	Export as PNG image and as DXF drawing for CAD programs
Opening track plans, saving plans with up to 50 tracks	

Pro features are marked in the program: in the tooltip as “Pro feature”, in the Android app with a padlock. If you choose a Pro feature in the Basic edition, a note explains what Pro offers and takes you straight to the licence dialog if you wish.

About this manual

> **New to this?**
> Then start with chapter 4 “First steps for beginners”. It explains the basic terms and leads you in ten steps to your first layout. You can look up the other chapters afterwards whenever you need a feature.

This manual describes the Windows version. The Android app and the web version work the same way; what is different there – mainly touch operation – is described in chapter 16, the web version in chapter 2. Ribbon commands are written as **Tab › Command**, for example **Tracks › Helix**. Keys look like this: Ctrl + S. The manual shows the English interface; how to switch the language is described in chapter 5.

Tip

Notes like this show shortcuts and proven ways of working.

Caution

Notes marked like this should be observed before building.

2 Installation

Windows

Requirements

- Windows 10 or Windows 11, 64-bit
- about 400 MB of free disk space
- screen of at least 1280 × 720 pixels, 1920 × 1080 recommended
- for the 3D view, a graphics card with DirectX support (present in every current PC)

No additional runtime is needed – everything TrackMaster Pro requires is included in the installation package.

Installing

1. Double-click the file **TrackMasterPro-1.5.0.msi**.
2. If Windows SmartScreen reports “Windows protected your PC”: click **More info** and then **Run anyway**.
3. Follow the setup wizard: accept the licence terms and choose the destination folder (default *C:\Program Files\TrackMaster Pro*).
4. Start the installation. Windows asks for permission to make changes – confirm with **Yes**.
5. After a few seconds the installation is complete. Close the wizard.

You will then find TrackMaster Pro in the **Start menu** and as a shortcut on the **desktop**. The manual is also in the Start menu.

Updating and uninstalling

Simply install a new version over the existing one: the installer detects the older version and replaces it. Your plans and your licence are kept. To remove the program, open **Settings › Apps › Installed apps**, select **TrackMaster Pro** and click **Uninstall**. Your plan files and the licence file are not deleted.

Silent installation for administrators

The package can be deployed to several computers without prompts:

Command	Effect
`msiexec /i TrackMasterPro-1.5.0.msi /qn`	installation without user interface
`msiexec /i TrackMasterPro-1.5.0.msi INSTALLFOLDER="D:\Programs\TrackMaster Pro" /qn`	installation into another folder
`msiexec /x TrackMasterPro-1.5.0.msi /qn`	uninstallation without user interface

Android

Requirements

- smartphone or tablet with Android 8.0 or later; a tablet of 10 inches or more is particularly pleasant for planning
- about 100 MB of free storage

Installing

1. Copy the file **TrackMasterPro-1.5.0.apk** to the device – via USB cable, cloud storage or as an e-mail attachment.
2. Tap the file on the device, for example in the **Files** app in the **Downloads** folder.
3. The first time, Android asks whether this app may **install unknown apps**. Tap **Settings**, allow it and go back.
4. Confirm with **Install**. TrackMaster Pro then appears in the app overview.

Install an update the same way over the existing app; your current plan is kept.

New versions

At start-up the program briefly checks the download page for a new version and reports it in the status bar. Under **File › Updates…** you can check yourself at any time and open the download page with installer, app and manual if you wish. Nothing is ever installed automatically; without an internet connection the check is simply skipped.

In the browser (web version)

TrackMaster Pro also runs without installation in a current browser (Edge, Chrome, Firefox or Safari), on a PC as well as on a tablet. The web version has the same tabs and features as the Windows program. On first use it loads a few megabytes and starts after a few seconds; after that it comes from the browser cache.

- **Plans:** the current plan is saved continuously in the browser and is there again on the next visit. **File › Save** downloads it as a **.mpr** file. **File › Save as…** (Ctrl + Shift + S) asks for the file name; in Chrome and Edge you also choose the folder, and **Save** then writes straight into that file. **File › Open** loads a **.mpr** or **.trackmaster.json** file. The content is the same as on Windows and Android: there you open a .mpr file by choosing “All files” in the Open dialog.
- **Licence:** under **File › Licence…** choose the licence file or paste its content. It applies only in this browser; after the site data has been deleted it has to be installed again.
- **3D view:** needs WebGL. If the browser reports that it cannot show 3D graphics, switch on hardware acceleration.
- **Layout suggestion:** Only in the web version, **Tracks › Layout suggestion…** builds small layouts from your track stock (see Layout suggestion in chapter 8).
- **Printing and exports** are downloaded as files or output via the browser's print dialog.

If the web version is embedded in a website, the **Home** button at the top left of the title bar leads back to that website's home page.

3 Licence and activation

The Pro edition is activated with a personal licence file. The file is digitally signed and cannot be altered; the program checks it at every start, without an internet connection.

Installing the licence

1. Save the licence file you received (extension **.lic**) on the computer, for example in the *Downloads* folder.
2. Open **File › Licence…** or **Home › Licence**.
3. Click **Install licence…** and select the file.
4. The dialog now shows **Pro**, the name of the licensee and, if applicable, the expiry date. The indicator at the bottom left of the status bar changes from “Basic” to “Pro”.

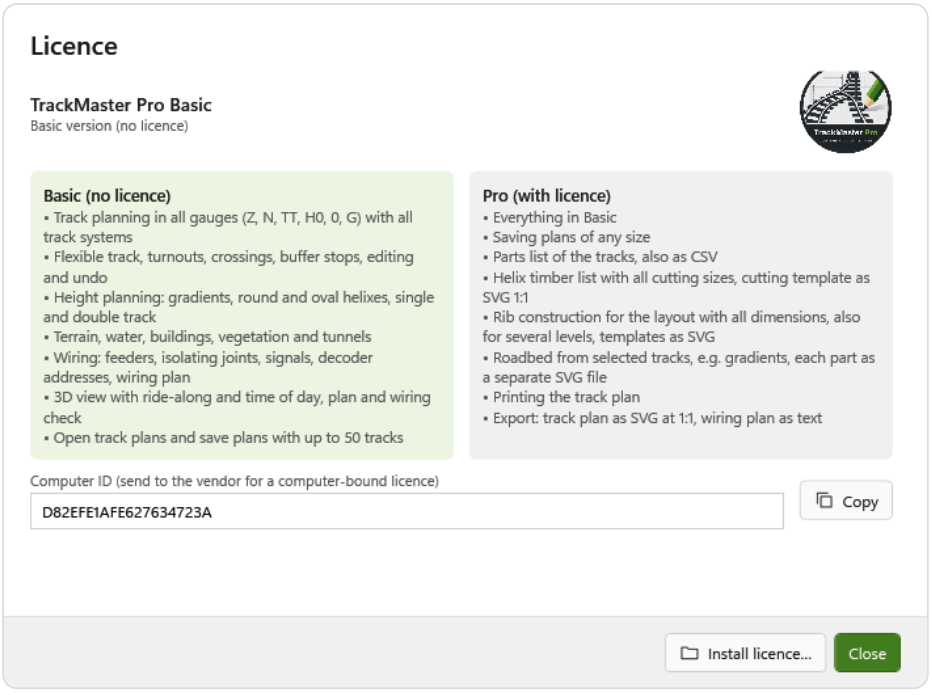


Figure 1 – The licence dialog shows the edition and what Basic and Pro contain.

The licence is stored in the user data (*%APPDATA%\TrackMaster Pro\trackmaster.lic*) and applies from now on at every start. On Android you install the licence the same way via **File › Licence**; you choose the file from the device storage.

Online activation (up to two devices)

A licence that is not bound to a computer may run on **two devices** at the same time – for example on the computer at home and on a tablet, on Windows, Android or in the browser. On the first start with the licence the device registers once over the internet; after that it checks about once a week that the registration still exists. The licence dialog shows the status in the **Activation** section.

- **Without internet** Pro keeps running for up to 14 days after installing the licence. A device that has been activated once stays activated without internet.

- If **both places are in use**, a third device stays on the basic edition. The message names the devices in use.
- To **change devices**, choose **Release this device** in the licence dialog on the old device. Then click **Activate now** on the new device. If the old device is no longer available, the publisher releases the place.
- In the browser every browser counts as a device of its own. If the site data is deleted, it registers as a new device.

Computer-bound licence

A licence can be bound to a particular computer or device. For this the publisher needs the **machine ID** (on Android: **device ID**). It is shown in the licence dialog; with **Copy** you put it on the clipboard and can paste it into an e-mail.

When a Pro feature is chosen

In the Basic edition, a note appears when you save plans with more than 50 tracks, and for the parts list, timber list, printing and the exports. With **Licence…** you open the licence dialog directly. Your plan stays open unchanged.

> **Caution**
> In the Basic edition, plans with more than 50 tracks cannot be saved. When you close the program, start a new plan or open another one, TrackMaster Pro asks whether the changes should be discarded.

4 First steps for beginners

Have you never planned a model railway on a computer before? Then this chapter is for you. It explains the most important terms, how to use the mouse and keyboard, and leads you step by step to your first small layout. You need no previous knowledge and no licence – everything works in the Basic edition.

The most important terms

A track plan is a drawing of your layout seen from above, at the correct scale. It is put together from individual **track pieces** – just like the real layout is built from the track you buy. You will meet these words all the time:

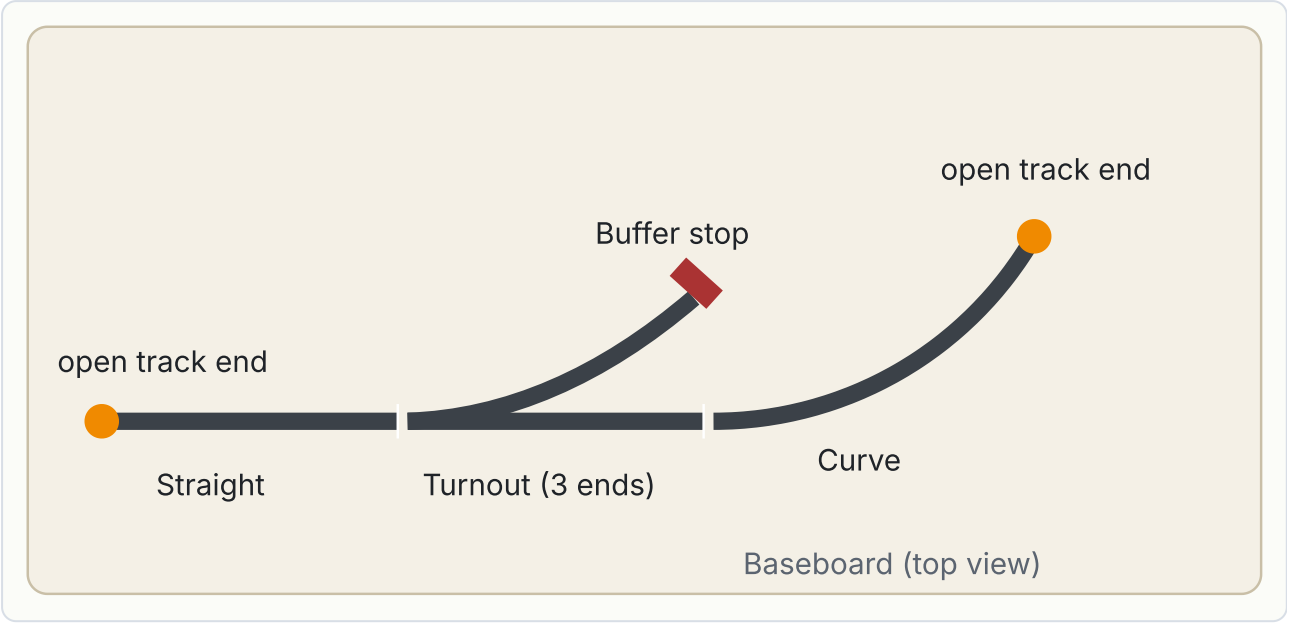


Figure 2 – Straight, turnout, curve and buffer stop; orange dots are open track ends.

Term	What it means
Gauge	The size of the model railway, for example H0 (scale 1:87, the most common), N (1:160, smaller) or Z (1:220, very small). You choose the gauge at the start; it determines which tracks are available.
Track system	A track range from one manufacturer. Pieces of one system fit together exactly. Choose the system you want to buy or already own.
Track piece	A single part: **straight**, **curve** (curved track), **turnout** (where the track divides), **crossing** or **buffer stop** (the end of a siding).
Track end	Every track piece has ends where the next one connects – a straight has two, a turnout three. An end without a connection is called an **open track end** and is marked orange in the plan.

Radius	How tight a curve is, measured from the centre of the circle to the middle of the track. Small radii save space; large ones look better and long coaches run more reliably.
Baseboard	The surface the layout is built on – often a wooden board, for example 2 × 1 m.
Millimetres	All dimensions in the program are in millimetres: 1 m = 1000 mm, 1 cm = 10 mm. A board of 2 × 1 m is therefore 2000 × 1000 mm.

More terms are explained in the Glossary at the end of the manual.

Mouse and keyboard – what you need

Action	How to do it
Click	Press the **left** mouse button once briefly. This selects tracks and presses buttons.
Drag	Hold down the left mouse button, move the mouse, then let go. This is how you move tracks.
Move the view	Hold down the **right** mouse button and move the mouse – the plan moves with it.
Zoom in and out	Turn the **mouse wheel**. The `0` (zero) key shows the whole layout again.
Undo	`Ctrl` + `Z` – hold down `Ctrl` and press `Z` – takes back the last step. As often as you like.
Cancel	`Esc` (top left on the keyboard) ends a tool or closes a dialog without changes.
Delete	Click a track and press `Del`.

> **Don't be afraid of mistakes**
> You cannot break anything. Every step can be undone with `Ctrl` + `Z`, and **Home › History…** even jumps back 20 steps at once. Feel free to try everything.

Where do I find what?

All commands are in the **ribbon** at the top. It is divided into **tabs**: **Home**, **Tracks**, **Scenery**, **Room**, **Wiring**, **View** and **3D**. Clicking a tab shows its buttons. On the far left is the **File** button with New, Open, Save and Print. When the manual says **Tracks › Gradient…**, it means: click the **Tracks** tab, then the **Gradient…** button.

On the left is the **track library** with all track pieces, in the middle the **drawing area** with your plan and on the right the **properties**: they show everything about what you have just clicked. Move the mouse over a button and wait a moment – a small note (tooltip) explains what it does. More about the window is in chapter 5.

Your first layout in 10 steps

We build a simple **oval** with a **siding** on a 2 × 1 m board in H0 gauge. It takes about a quarter of an hour.

1. **Create a new plan.** Click **File** and then **New** (or press Ctrl + N). The **New plan** window opens.
2. **Choose gauge and baseboard.** Under **Gauge** choose **H0**, as the shape **Rectangle**. Enter **2000** as the width and **1000** as the depth (millimetres!). Click **Create plan**. An empty baseboard now lies on the drawing area.
3. **Choose the track system.** At the top of the track library on the left, choose the track system. The list below now shows all track pieces with number and size. With the buttons **Straight**, **Curves**, **Turnouts** etc. above it you see only that type.
4. **Place the first curve.** Click **Curves** and then once on a curve with a small radius. It appears on the board. Its free end is **orange**: that is where the next track goes.
5. **Complete the semicircle.** Keep clicking the **same curve**. Each click attaches a piece to the orange end until a semicircle is formed (with 30° curves that is six pieces; the angle is shown with the track in the library). It is quicker with the Enter key: it attaches the last chosen track again each time.
6. **Attach straights.** Click **Straight** and two or three times on a straight. The line now continues straight ahead.
7. **Build the second half.** Place six curves again and then as many straights as before. When the oval closes, the program connects the two ends by itself. The status bar at the bottom then shows **Open ends: 0**.
8. **Add a siding.** Click a straight in the oval and delete it with Del. Two orange ends appear. Click one of them, then a turnout under **Turnouts** – it is inserted there. Close the gap with a straight; if it does not fit exactly, click **Tracks › Close gap**. Now click the free end at the branch of the turnout, place two straights and finally a **buffer stop** under **Other**.
9. **Check.** Press F7 (or **Home › Collision check**). The list reports whether tracks touch, project over the edge or ends are open. A click on an entry shows the spot. If the list is empty, everything is fine.
10. **Look and save.** Click **3D › 3D view** and turn the layout with the left mouse button held down. Close the window and save with Ctrl + S; enter a file name. Done – your first layout!

> **Tip**
> If the turnout faces the wrong way, press A (change connector) or F (mirror) – this turns the last track placed. If a track lies off to one side, drag it with the mouse to an orange end: it snaps on by itself.

When something doesn't work

What happens	What helps

The track appears in the wrong place.	First click the orange end where you want to continue, then click the track in the library.
I can't see my plan any more.	Press `0` – the whole layout is shown again.
A wrong track was placed.	`Ctrl`+`Z`, or click the track and press `Del`.
The oval does not quite close.	Try a different straight or bridge the gap with **Tracks › Close gap** (chapter 8).
A tool “sticks”, every click places something.	Press `Esc` – after that a click only selects again.
A track cannot be moved.	It is locked. Untick **locked** in the properties.

Where next?

Simply read the other chapters when you need a feature:

- Plan an L- or U-shaped board, modules or timber benchwork: chapter 7.
- More about tracks, flexible track, parallel tracks and aligning: chapter 8.
- Let lines climb, build a second level or a helix: chapter 9.
- Mountains, lakes, houses and trees: chapter 10. Fit the layout into your room: chapter 11.
- Power, signals and digital control: chapter 12.
- Shopping list with prices, printing at full size: chapter 15.

The **sample plan** (**File › Load sample plan**) shows a finished layout with station, scenery and signals. Have a look at it and click on individual parts – the properties on the right reveal how everything is done.

5 The workspace

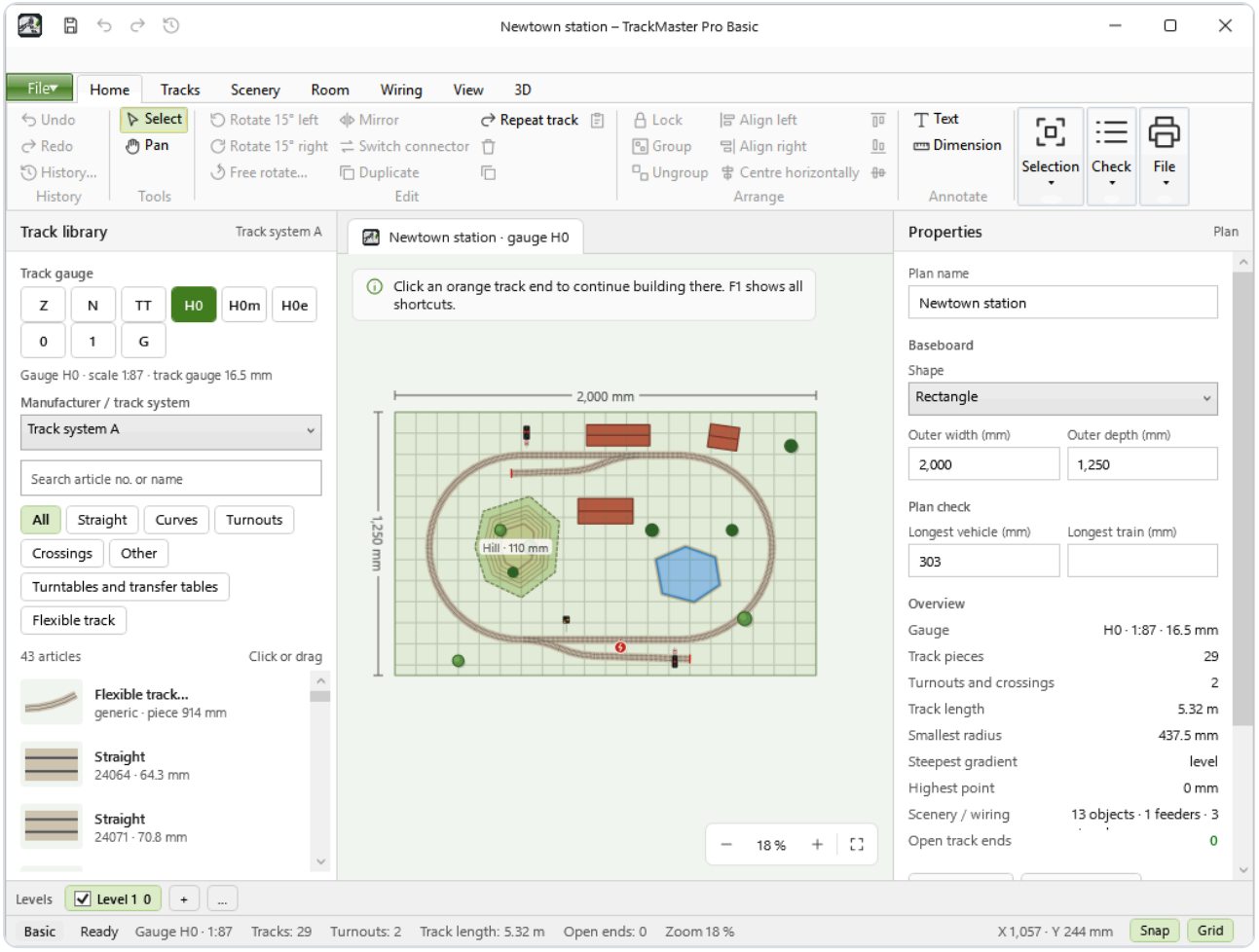


Figure 3 – The main window with the sample plan “Newtown station”.

The main window is divided into six areas:

Area	Purpose
Ribbon (top)	All commands, arranged in the tabs Home, Tracks, Scenery, Room, Wiring, View and 3D. Via **File** you reach New, Open, Save, parts list, export, printing, sample plan, licence and language. The quick access toolbar at the very top left contains Save, Undo, Redo and History.
Track library (left)	Choose gauge and track system, search for tracks and filter by type (straight, curves, turnouts, crossings, other). A click attaches the track, dragging places it freely.
Drawing area (centre)	The track plan on the baseboard, with dimension chains, grid and the open track ends. The zoom can be set at the bottom right.
Properties (right)	Shows what is currently selected: the plan with baseboard dimensions and overview, a track with position, height and digital address, or an object such as a building, signal or feeder. Values can be entered directly.

Level bar (bottom)	All levels of the plan with name and base height: the tick shows or hides a level, a click makes it the active level, **+** adds a new one above, **…** opens the Levels dialog (chapter 9).
Status bar (bottom)	Edition, gauge, number of tracks and turnouts, track length, open ends, zoom and mouse position in millimetres; plus the switches for snapping and grid.

Moving the view

- **Zoom:** mouse wheel, the keys + and − or the zoom bar at the bottom right.
- **Pan:** hold the right or middle mouse button and drag – or hold Space and drag with the left mouse button. Alternatively use the tool **Home › Pan**.
- **Show the whole layout:** key 0 or **View › Whole layout**.

Rotating the view freely

The track plan can be rotated on screen to any angle – for example so that the layout lies the way you will later stand in front of it. Only the view is rotated: tracks, dimensions, parts list and printout remain unchanged.

- **With the mouse:** hold Ctrl and turn the mouse wheel – in 5° steps, with Shift as well in 1° steps.
- **Via the ribbon: View › Rotate** with **15° left**, **15° right** and **Rotate view…** for an exact angle.
- **Back:** while the view is rotated, a compass at the top right shows the plan's north direction and the angle. A click on it – or **View › Straighten** – straightens the view again.

The dimension chains at the baseboard rotate along; their labels always stay readable.

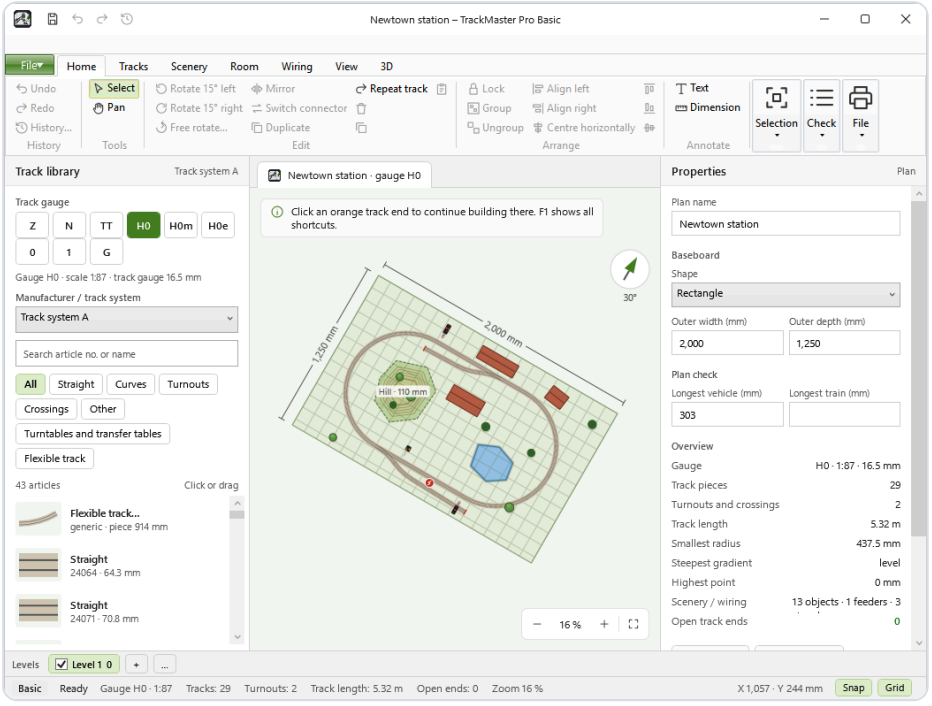


Figure 4 – View rotated by 30° with the compass at the top right.

Design and font

Under **View › Design and font** (also in the menu **File › Appearance…**) you choose the look of the program. The choice is saved and also applies at the next start. After **Apply** the window is briefly rebuilt; the open plan is kept.

Design	Appearance
Logo green	Default: green and anthracite from the logo, light workspace.
Classic blue	Blue accents, light workspace.
Blueprint	The plan as a technical drawing: white tracks on a blue background.
High contrast	Black tracks on a white baseboard – easy to read, even in bright light.
Workshop	Warm wood tones: baseboard like plywood, accents in copper.

Seven **fonts** are available – from the standard font and a narrow technical font to a serif font and a typewriter font. The **font size** can be set in four steps (Small, Normal, Large, Very large); a preview in the dialog shows the effect immediately.

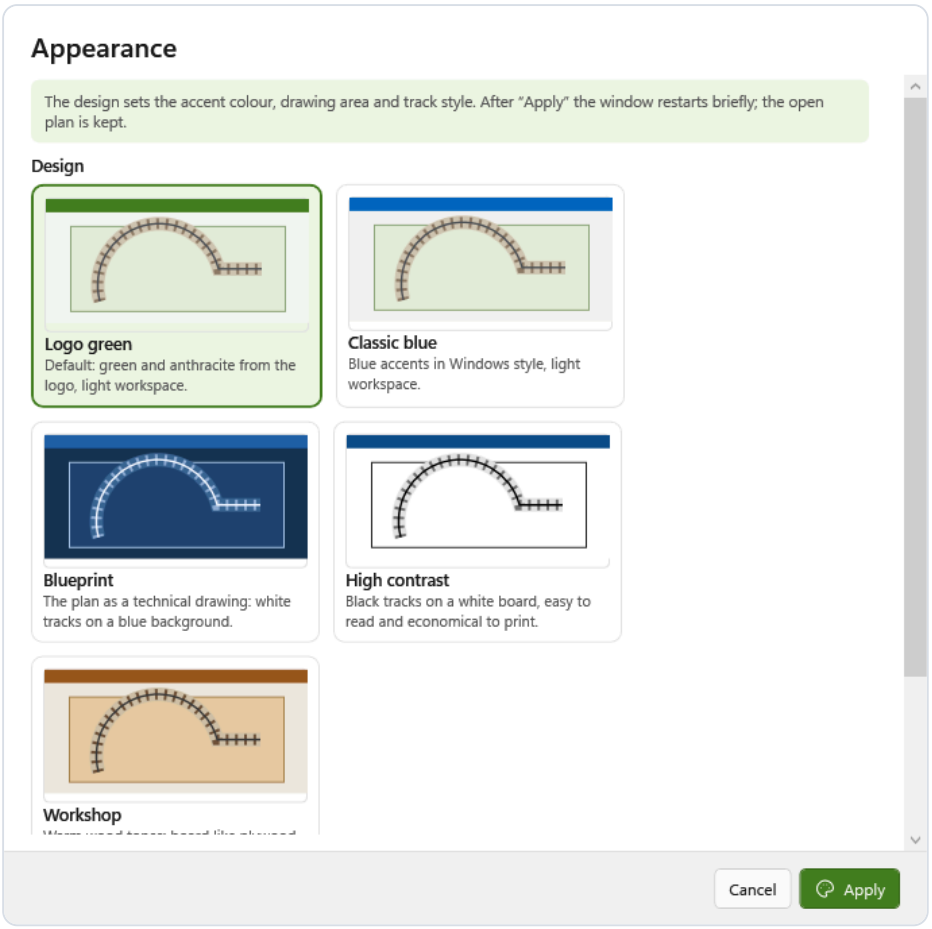


Figure 5 – Appearance dialog with designs, font and font size.

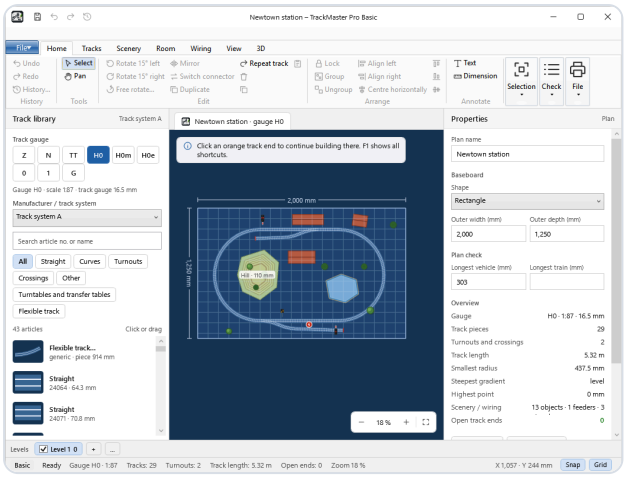


Figure 6 – Blueprint design.

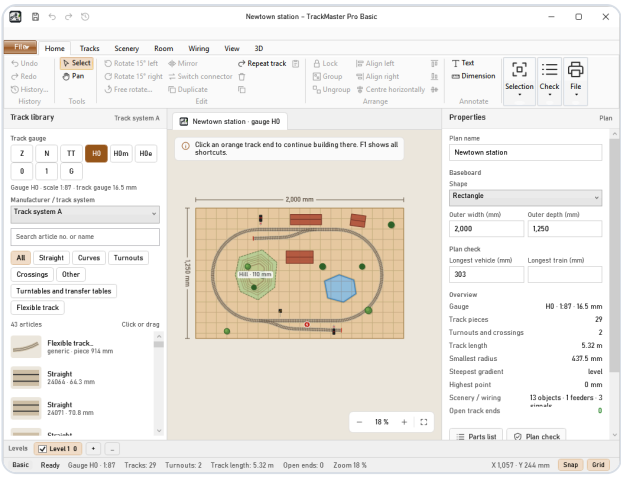


Figure 7 – Workshop design with a narrow font.

Language

The interface is available in German and English. At the first start it follows the system language. Under **File › Sprache / Language…** you switch it; on Windows the window then restarts and the open plan is kept, on Android the new language applies from the next start of the app. Plans do not depend on the language.

View tab

In the **View** tab you switch on what the drawing area shows: **Grid**, **Track ends**, **Article numbers** on every track, **Heights** (top of rail at the track ends) and **Scenery**. There are also the groups **Rotate** and **Appearance** (see above) and **About** with version and edition. **Controls and keyboard shortcuts** (F1) shows all mouse and keyboard commands at a glance.

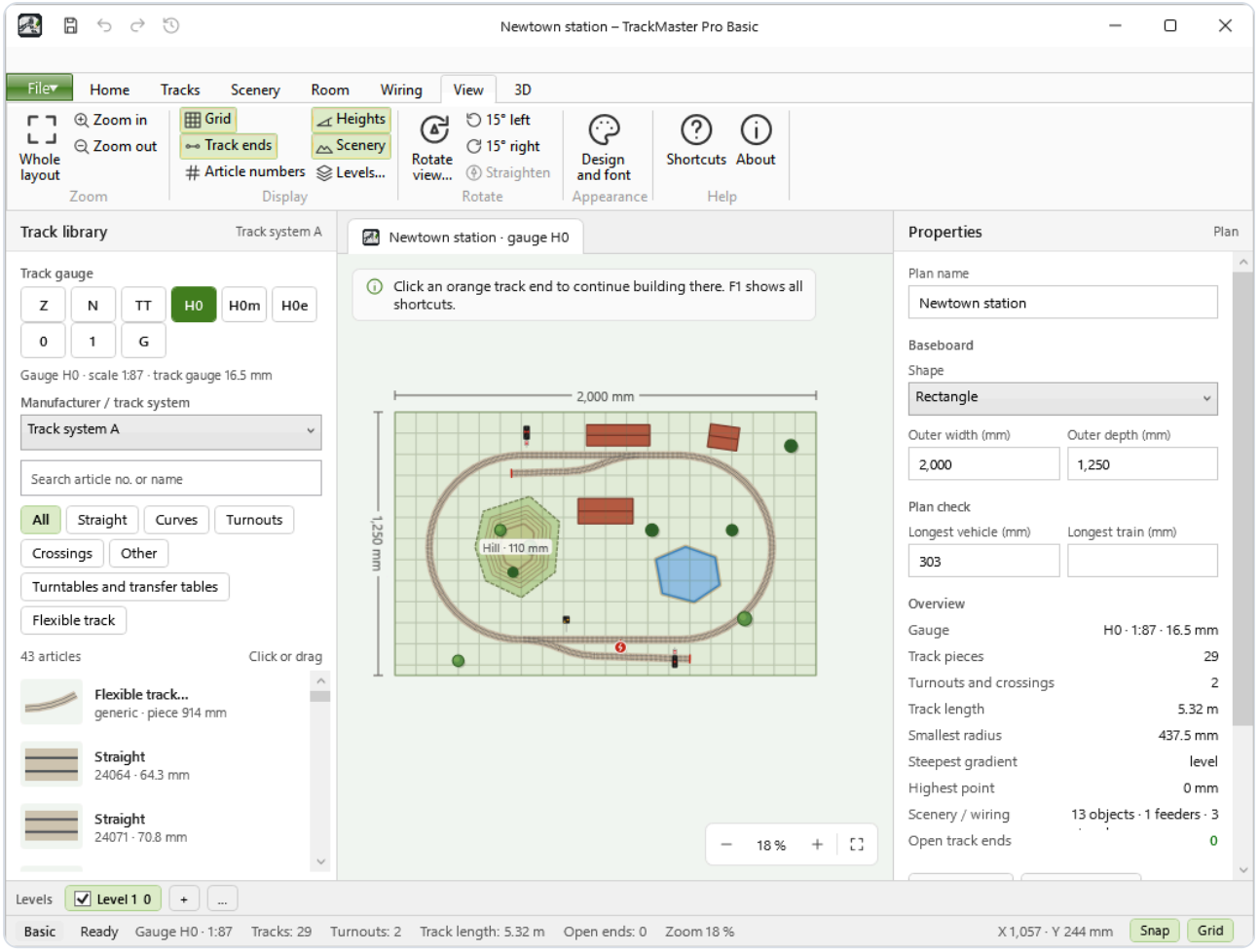


Figure 8 – View tab.

6 Creating a plan and attaching tracks

This chapter summarises the basic moves and shows all baseboard shapes. A detailed guide for getting started is in chapter 4.

Creating a new plan

1. Choose **File › New** (Ctrl+N).
2. Choose the **gauge**.
3. Choose the **shape of the baseboard** and enter the dimensions. The preview shows the shape immediately.
4. Confirm with **Create plan**.
5. Then enter the name of the plan in the properties under **Plan name**.

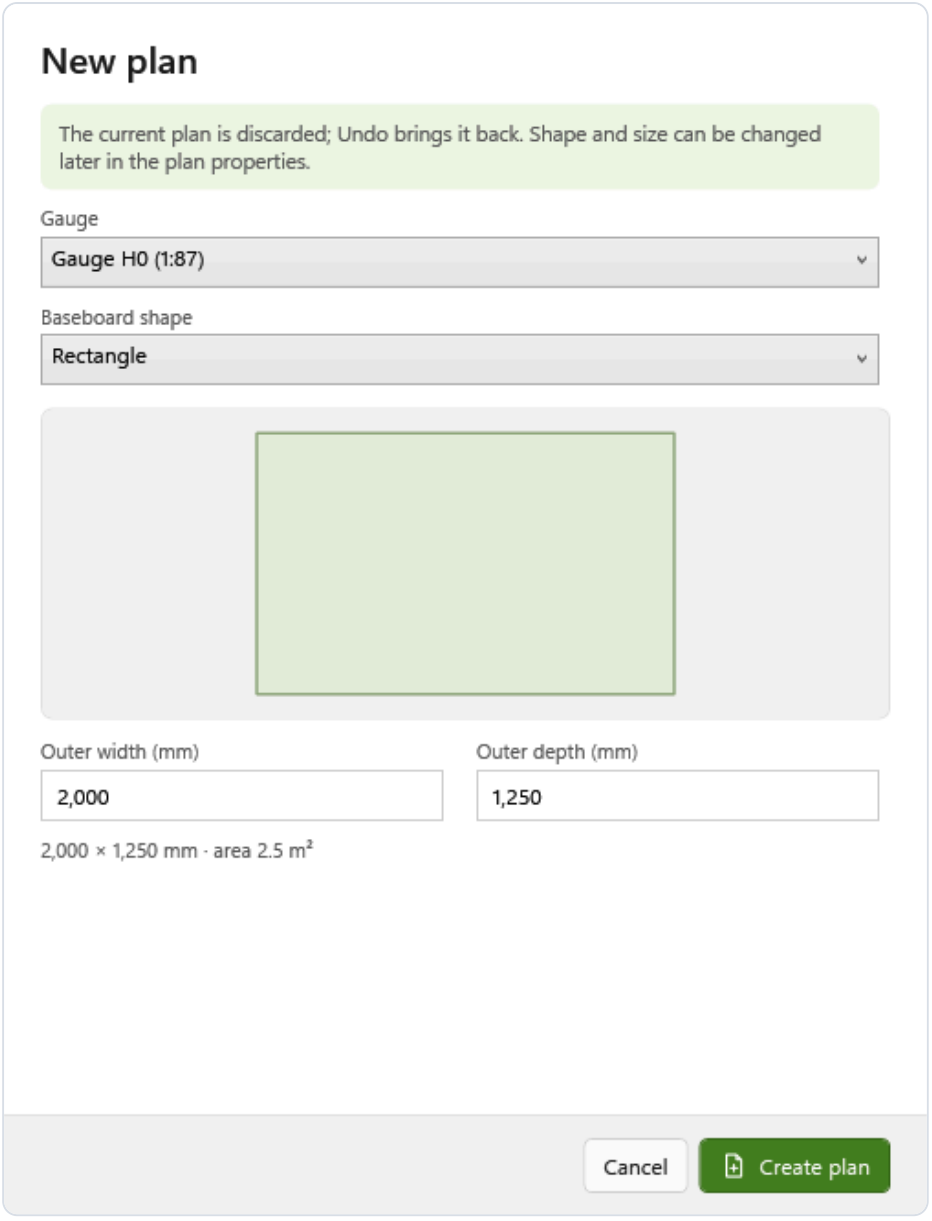


Figure 9 – New plan: gauge, baseboard shape and dimensions.

Baseboard shape	Dimensions and particulars
Rectangle	width and depth
L shape	outer dimensions and leg depth, orientation in 90° steps
U shape	outer dimensions and leg depth; the two legs may have different lengths
T and Z shape	outer dimensions, leg depth and orientation
E and G shape	outer dimensions, leg depth and orientation; the E shape has three legs, the G shape an inward hook
Ring	layout all round with an open centre, for example for an access hatch in the middle
Modules	layout made of individual rectangular modules (chapter 7)

Shape and dimensions can be changed at any time later: click a free spot of the drawing area or choose **Scenery › Baseboard**; the properties then show the baseboard dimensions. More about the baseboard, modules and benchwork is in chapter 7.

Choosing a track system

In the track library, choose the track system you want to build with under **Manufacturer / track system**. The library then shows all track pieces of this system with article number and dimensions. You can also mix several systems in one plan.

Attaching tracks

1. Click a curved track in the library. It appears in the middle of the baseboard; its free end is marked **orange** – this is the **active track end**.
2. Keep clicking curves. Each new track is attached to the active end until the semicircle is complete.
3. Add straights and the second semicircle. When the oval closes, TrackMaster Pro connects the ends automatically.
4. Click the orange end of another track to continue building there, or press `E` for the next open end.
5. Insert a turnout, run a parallel track and connect it again with a second turnout.

> **Tip**
> With `A` (switch connector) you lay the most recently placed track with its other connector – this quickly turns a right-hand turnout in the direction of travel into one that receives the track from the other side.

Checking the plan

Click **Home › Collision check** (`F7`). The list shows overlaps, spacing that is too tight and open ends; a click on an entry shows the spot in the plan (chapter 13).

Sample plan

Under **File › Load sample plan** the layout “Newtown station” opens, with scenery, wiring and signals. It is well suited for trying out all features.

7 Baseboard, modules and benchwork

The baseboard can be dragged with the mouse or typed in to the millimetre, divided into modules, rotated and given a timber benchwork.

Drag baseboard and type in dimensions

Switch on **Scenery › Drag baseboard**. The baseboard then shows handles:

- **Dragging corners** changes width and depth. Baseboards and modules are at least 50 × 50 mm.
- **Round handles** on L, U, T, Z, E, G shape and ring: at the inner edge dragging changes the depth of the leg, at the free end its length.
- **Dragging a module** moves it together with its tracks and terrain; its edges snap to the neighbouring modules.
- **Clicking without dragging** opens the dimensions for typing in: for a module its name, position (X, Y of the top left corner), width and depth, otherwise width, depth and leg depth of the baseboard. A click on a round handle opens **Individual legs...**.

Esc ends the tool. Every step can be undone with Ctrl+Z.

Individual legs

On L, U, T, Z, E, G shape and ring the legs initially share one depth. Under **Individual legs...** (in the baseboard dimensions or in the baseboard properties) each leg gets its own **depth** and, where it fits, its own **length**. Empty fields take the common leg depth or the full length; **Reset all** restores the initial state.

Rotate layout

Scenery › Rotate layout 90° rotates the baseboard together with tracks, terrain and objects by 90° clockwise; rotating four times gives the initial state again. In the dimensions dialog of a module, **Rotate 90°** rotates only that module with its content. To rotate only the view, see chapter 5; free rotation in all directions is available in the 3D view (chapter 14).

Layout made of modules

Under **Scenery › Modules...** the layout is assembled from rectangular modules. An existing baseboard is split into modules, one per rectangle of the shape. The table shows each module's name, position (X, Y) and size in millimetres.

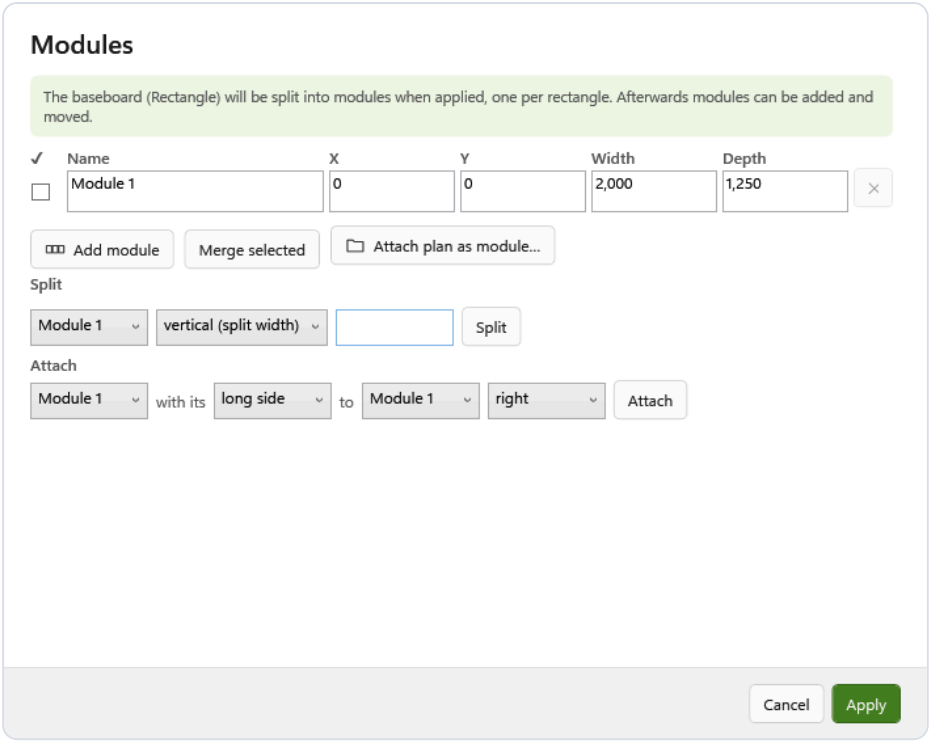


Figure 10 – Modules dialog with split and attach.

Command	Effect
Add module	new module to the right of the layout
Merge selected	joins the ticked modules into one if together they form a rectangle
Split	divides a module **vertically** (split width) or **horizontally** (split depth) into two modules – at a position entered in mm or, with an empty field, in the middle. The parts are called "… a" and "… b".
Attach	places a module **with its long or short side** flush to the right, bottom, left or top of another one; it is rotated by 90° if necessary
Attach plan as module…	places a saved plan as a new module next to the layout, with all its tracks and terrain
×	removes the module

Only **Apply** changes the plan. Tracks and terrain move with their module. The plan check reports tracks that run across a module joint.

Height Z (web version): In the column **Height Z** you set a module higher or lower than the others, e.g. a station module 80 mm above the line or a harbour basin 40 mm lower (negative value). The tracks on the module move along, newly laid tracks lie on its surface, and the 3D view shows the module at its height. Bridge the difference to the neighbouring module with a gradient (**Tracks › Gradient…**); until then the plan check shows the transition as an open track end. The size dialog of a single module (**Scenery › Drag baseboard**, click a module) also has the Height Z field. The column **Thickness** or the field **Baseboard thickness** sets the thickness of the module board (empty = default); in the 3D view a raised module appears as a board of this thickness.

Cutting tracks at the module edge (web version). Right-click a track that runs across a module edge and choose **Cut at module edge**. The program splits the track exactly at the edge into two flex track pieces. Position, height and gradient stay the same; feeders and isolating joints move along. The same function is available under **Scenery › Cut at module edge**: it splits the selected tracks, or all tracks across module edges if nothing is selected. Turnouts, crossings and turntables cannot be split; they must lie entirely on one module.

Timber benchwork

Scenery › Benchwork… plans the timber frame under the baseboard: aprons all round and cross members under each baseboard part, legs in the corners and along the long sides, and if you wish foot braces, levelling feet and a top board.

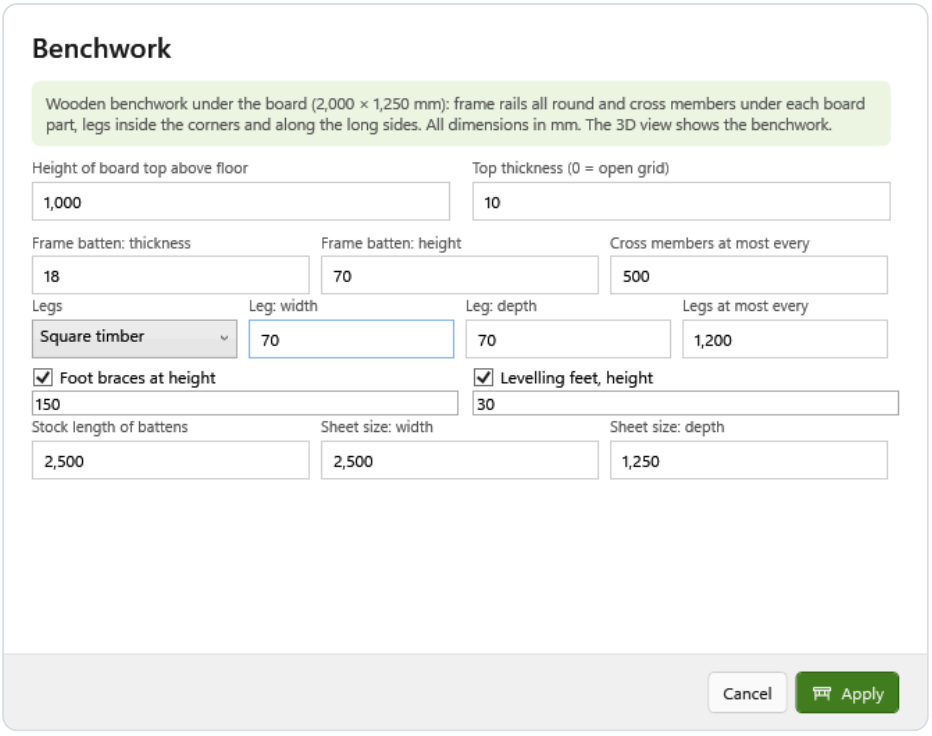


Figure 11 – Benchwork dimensions.

Setting	Meaning
Height of board top above floor	working height of the layout
Top thickness	thickness of the top board; 0 gives an open grid for rib construction
Frame batten	thickness and height of aprons and cross members; cross members at most every … mm
Legs	square timber or L-leg from two battens, width and depth, spacing at most … mm
Foot braces, levelling feet	height of the foot braces and of the levelling feet (M8 with tee nut)
Stock length, sheet size	trade sizes the material list calculates with

The **material list** names battens and square timber by trade length, sheets, screws and glue. The **cutting list** shows each part with cross-section, length and quantity; both can be copied or saved as a CSV file. Parts that are too long and top boards that are too large are reported with a hint on how to join them. The 3D view shows the benchwork under the baseboard. **Remove** in the dialog takes it out of the plan again.

8 Laying and editing track

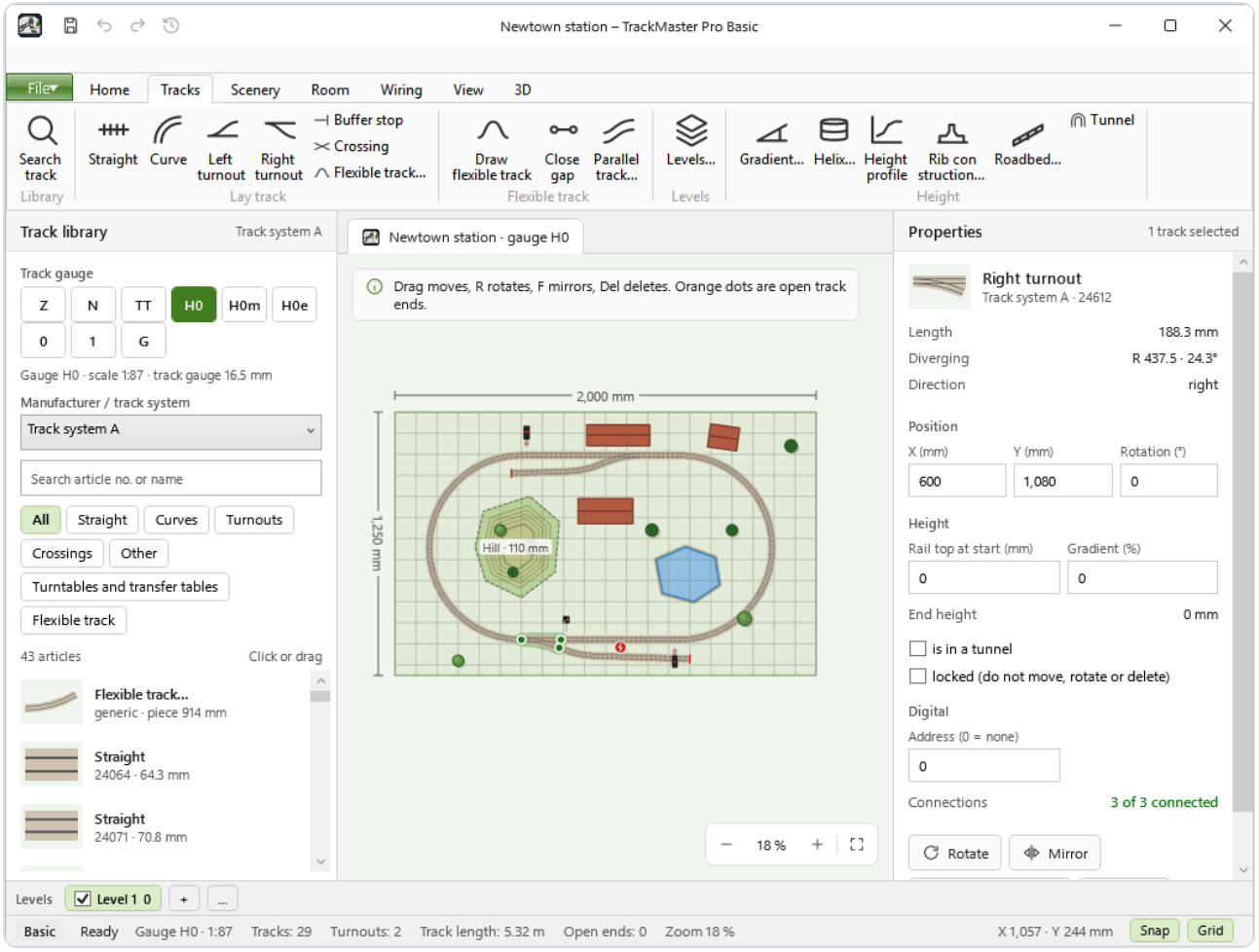


Figure 12 – Tracks tab with a selected turnout; the properties show its dimensions, height and digital address.

Track library

- Choose the **gauge** at the top (Z, N, TT, H0, 0, G and further gauges in the list).
- Choose the **track system** in the drop-down list.
- **Search** by article number or name; **Tracks › Search track** jumps straight into the search field.
- **Filter** by All, Straight, Curves, Turnouts, Crossings and Other.
- **Clicking** attaches the track to the active end, **dragging** places it freely on the baseboard; near a free end it snaps on.

The group **Tracks › Lay track** offers quick commands for **Straight**, **Curve**, **Left turnout**, **Right turnout**, **Buffer stop** and **Crossing** of the selected system.

Turntables are available for every gauge and every track system; you find them in the library under the filter **Turntables and transfer tables** (or search for “turntable”). They are attached like tracks; every track exit is a track end of its own. In the web version **Tracks › Turntable** attaches the turntable of the selected system directly at the active track end. Where the manufacturer offers no turntable of its own, a matching one of another manufacturer or a generic turntable (“generic”) is

provided – its dimensions are preliminary and should be checked against the real thing before building.

> **Note on dimensions**
> The track geometry follows the catalogue data. A few track systems are stored with provisional dimensions and marked in the library; check these against the real product before building.

Selecting and moving

- **Click** on a track selects it, Shift+click adds more.
- **Dragging on a free area** draws a selection rectangle; Ctrl+A selects everything.
- **Dragging** moves the selection. If a free end comes close to another one, the whole group snaps on there. A single track snaps onto any free end, even if it lies the wrong way round – it is turned to fit.
- **Arrow keys** move by 1 mm, with Shift by 10 mm.
- Snapping can be switched off in the status bar with **Snap**.

When a track is selected, the properties show its **usable length** as well as position, rotation and height: the length of the track section up to the next turnout, crossing or track end – in other words, how long a train standing there may be. If a **longest train** is entered in the plan properties, the usable length turns red when the train does not fit.

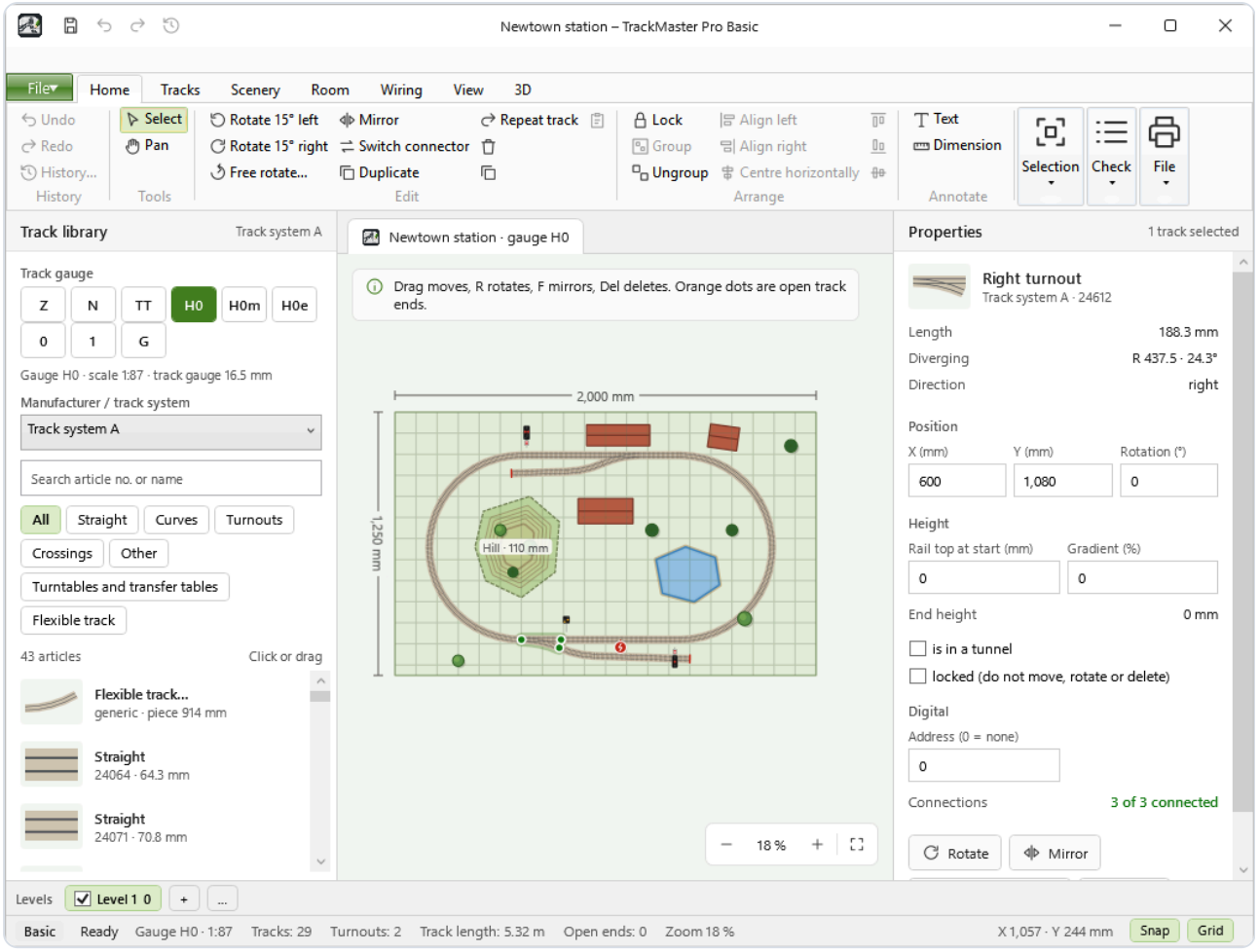


Figure 13 – Home tab with the Arrange group; the properties show the “locked” tick box.

Editing

Command	Key	Effect
Rotate 15° left / right	Shift + R / R	rotates the selection about its centre
Free rotate...	–	rotates the selection by any angle (quick choice ±15°, ±45°, ±90°)
Mirror	F	curves, turnouts and crossings reversed
Switch connector	A	lays the track with the next connector at the same spot
Repeat track	Enter or Space	attaches the last track again at the active track end – for example several flexible tracks, straights or curves in a row; a mirrored curve stays mirrored
Duplicate	Ctrl + D	copy of the selection
Copy / Paste	Ctrl + C / Ctrl + V	tracks including feeders, isolating joints and signals via the clipboard, also into another plan of the same gauge; pasted at the mouse position

Delete	Del	removes tracks or the selected object
Undo / Redo	Ctrl + Z / Ctrl + Y	step by step back or forward
History…	–	list of the last 20 steps in both directions; a click jumps straight to that state (**Home › History** or quick access toolbar)

The position (X, Y in millimetres) and rotation of a track can also be entered exactly in the properties.

Labels and dimension lines

In the group **Home › Annotate** you place **text** (click the spot and enter the text) and **dimension lines** (click two points; track ends and baseboard corners snap on). The program calculates the length itself. Change text height, rotation and text in the properties; labels and dimensions also appear in the printout.

Arrange: lock, group, align

The **Home › Arrange** group helps when the plan gets larger:

Command	Effect
Lock	Selected tracks can no longer be moved, rotated or deleted – ideal for finished areas that must not slip by accident. A second click unlocks them. You can also lock single tracks with the **locked** tick box in the properties.
Group	Combines the selected tracks into a group: a click on one of them then selects them all, and they move together – for example a whole station. **Ungroup** separates them again.
Align	Aligns several separate groups of track flush: **left**, **right**, **top**, **bottom** or **centred** horizontally or vertically. Select at least two groups that are not connected to each other.

When several tracks are selected, the same commands are also available as buttons in the properties. If the selection contains locked tracks, a note points this out.

The field **Height of all selected tracks (mm)** in the properties puts all selected tracks at one common height (rail top); any gradient is removed and locked tracks stay as they are. If the tracks lie at different heights the field is empty and the range is shown above it. For a ramp use **Gradient…**.

Parallel track

For a double-track line or a passing loop, TrackMaster Pro lays a second track at a fixed spacing next to existing tracks:

1. Select the tracks of the line (drag a rectangle or Shift+click). They should be connected.
2. Click **Tracks › Parallel track…**.

3. Enter the **track spacing** centre to centre – the usual spacing for the gauge is preset – and choose the **side**: left or right in the direction of travel.

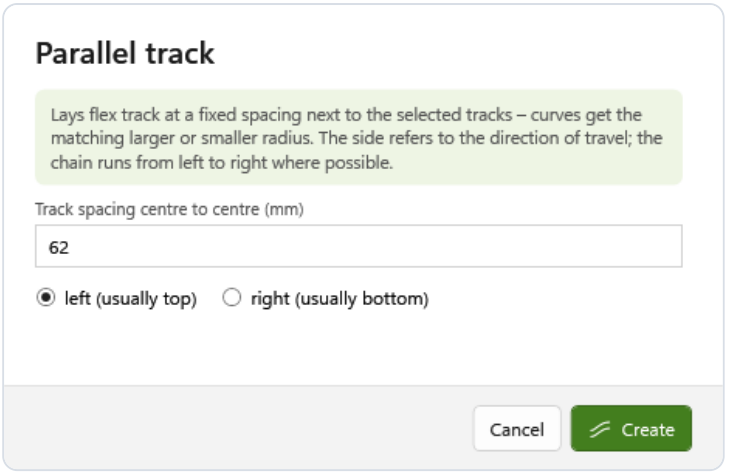


Figure 14 – Parallel track dialog.

The new track is laid as **flexible track** of the selected system; curves automatically get the matching larger or smaller radius. If it ends up on the wrong side, undo it with Ctrl + Z and choose the other side.

Layout suggestion from your own track

You have a box full of track but no idea for a layout? The layout suggestion builds small layouts from your quantities – not large layouts, but ideas to get started. *For now this feature is only available in the web version.*

1. Set up the **baseboard** (size and shape) and choose your **track system** in the track library.
2. Click **Tracks › Layout suggestion…**.
3. For each piece, enter **how many** you have. Empty or 0 means none. If you have already entered your stock in the bill of materials, it is filled in here.
4. Click **Calculate suggestions**. After a moment the suggestions appear with a small preview, the number of tracks and the size.
5. If you like one, click **Apply**. Ctrl + Z removes it again.

Depending on your stock, these shapes are offered: **oval**, **oval with siding** (or two), **oval with passing loop**, **double oval** (two loops one inside the other) and the **figure eight** with a crossing in the middle. Every suggestion is closed, fits the baseboard and passes the plan check; it never uses more pieces than you entered.

> **No suggestion?**
> An oval needs curves for a full circle of one radius (e.g. 12 curves of 30°), and the baseboard must be at least twice the radius deep plus a little margin. Sidings need a turnout, a matching counter curve and a buffer stop. If “Replace existing tracks” is ticked, the suggestion replaces the tracks in the plan; otherwise it is added.

Flexible track

Flexible track can be laid in three ways:

Draw flexible track

1. Switch on **Tracks › Draw flexible track**.
2. Put the mouse on an open track end and drag. The flexible track leaves the track tangentially and bends along.
3. If you come close to another track end, it snaps on: the flexible track is then laid as a matching double curve that joins both ends without a kink.
4. Release the mouse button. Length and smallest radius are shown while dragging.

Reshaping: with the tool switched on, grab a laid flexible track at one end and drag – the other end stays in place.

Close gap

Click a track end (active end), select the counterpart and click **Tracks › Close gap**. TrackMaster Pro connects both ends with a suitably bent flexible track.

Flexible track with dimensions

Via **Tracks › Flexible track…** or the entry “Flexible track…” in the library you create a straight piece of any length or a curve with any radius and angle.

If a track system has flexible tracks of its own, they are listed individually in the library. The parts list counts flexible track in whole lengths of the respective size, with 5 % waste.

> **Caution**
> When dragging, watch the smallest radius shown. It should not be smaller than the smallest curve radius your vehicles can negotiate.

9 Heights, levels, helixes and rib construction

Height of a track

Every track has a **top-of-rail height** at its start and a **gradient** in percent. You enter both in the properties; TrackMaster Pro calculates the height at the other end. With **View › Heights** the heights at the track ends appear in the plan; higher sections are always drawn above lower ones.

Gradient over a track sequence

1. Select the tracks of the ramp (draw a rectangle or Shift+click).
2. Click **Tracks › Gradient…**.
3. Enter the **start height** and choose under **Input** whether you give the **height at the end** (the gradient in percent is calculated from the length of the track sequence) or the **gradient in percent**; reverse the direction if needed. Below the fields you see the ramp length and the gradient or end height.

The gradient is applied evenly over all selected tracks. If two parallel track sequences are selected – such as the two tracks of a double-track ramp or helix – the inner, shorter track gets exactly the gradient entered; the outer one rises so that it arrives at the same height.

Gradient	Assessment
up to 2 %	close to the prototype, suitable for all trains
2 to 3 %	common on model railway layouts
over 3 %	the plan check warns; long trains need a second locomotive
over 5 %	the plan check reports an error

Height at a joint (web version). For selected tracks the heights at the joints are shown in the plan (↕ in mm); for raised lines also without a selection once you have zoomed in far enough. Double-click a joint – or right-click and choose **Height at joint…** – to change the height there. Both tracks at the joint get the new height, their other ends stay in place and the gradient adjusts; turnouts and crossings are raised as a whole.

Height profile

How high does the line lie at which point, and where is it too steep? Select connected tracks – for example a ramp or the whole line between two stations – and click **Tracks › Height profile**. The chart shows the height of the top of rail over the distance travelled. **Red** sections are steeper than recommended. This shows you at a glance where you should make changes.

Levels

A layout can have up to **10 levels**, for example for the fiddle yard, main level and mountain line. Each level has a name and a **base height**; every track and every terrain object belongs to exactly one level.

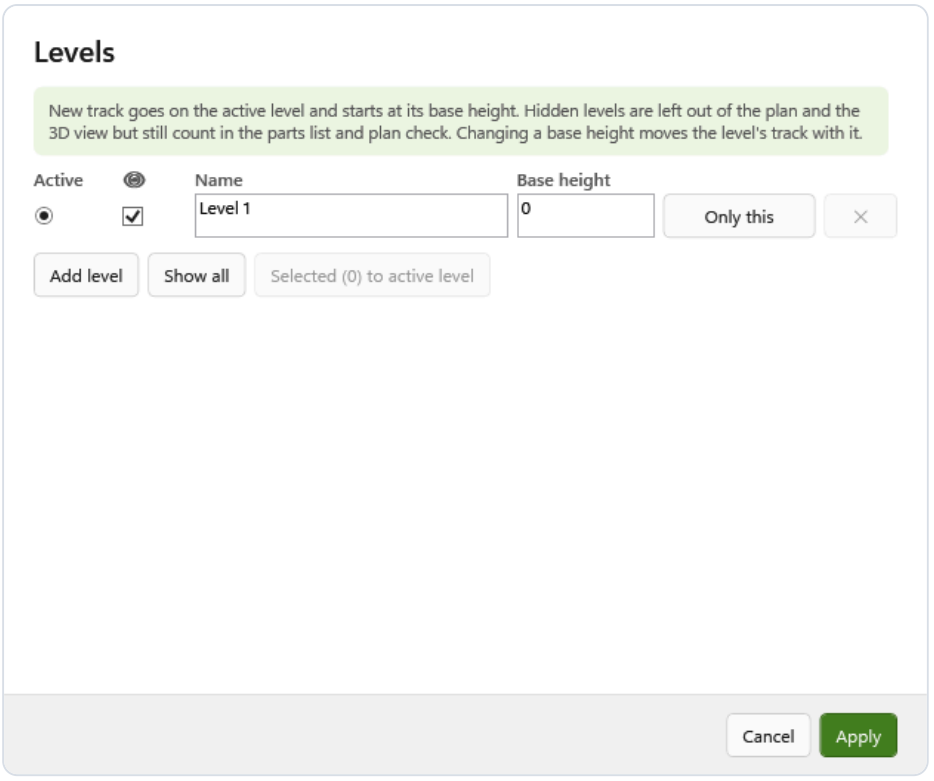


Figure 15 – Levels dialog.

- **Level bar** at the bottom of the window: the tick shows or hides the level, a click makes it active, **+** adds a new level above.
- **Levels dialog** (**Tracks › Levels…** or **…** in the level bar): add, rename, change base height, remove levels (their tracks go to the level below), show **Only this** or **Show all**.
- **New tracks** go onto the active level and start at its base height. **Selected to active level** moves selected tracks; their height changes by the difference in height between the levels.
- If the base height of a level changes, its tracks move with it. **Hills** of a level stand on its base height, so the terrain adapts automatically.
- Hidden levels are missing from the plan and the 3D view, but still count in the parts list and plan check.

Helix

A helix overcomes large differences in height on a small area. TrackMaster Pro builds it from curved tracks of the catalogue – **round** or **oval**, **single-track** or **double-track**.

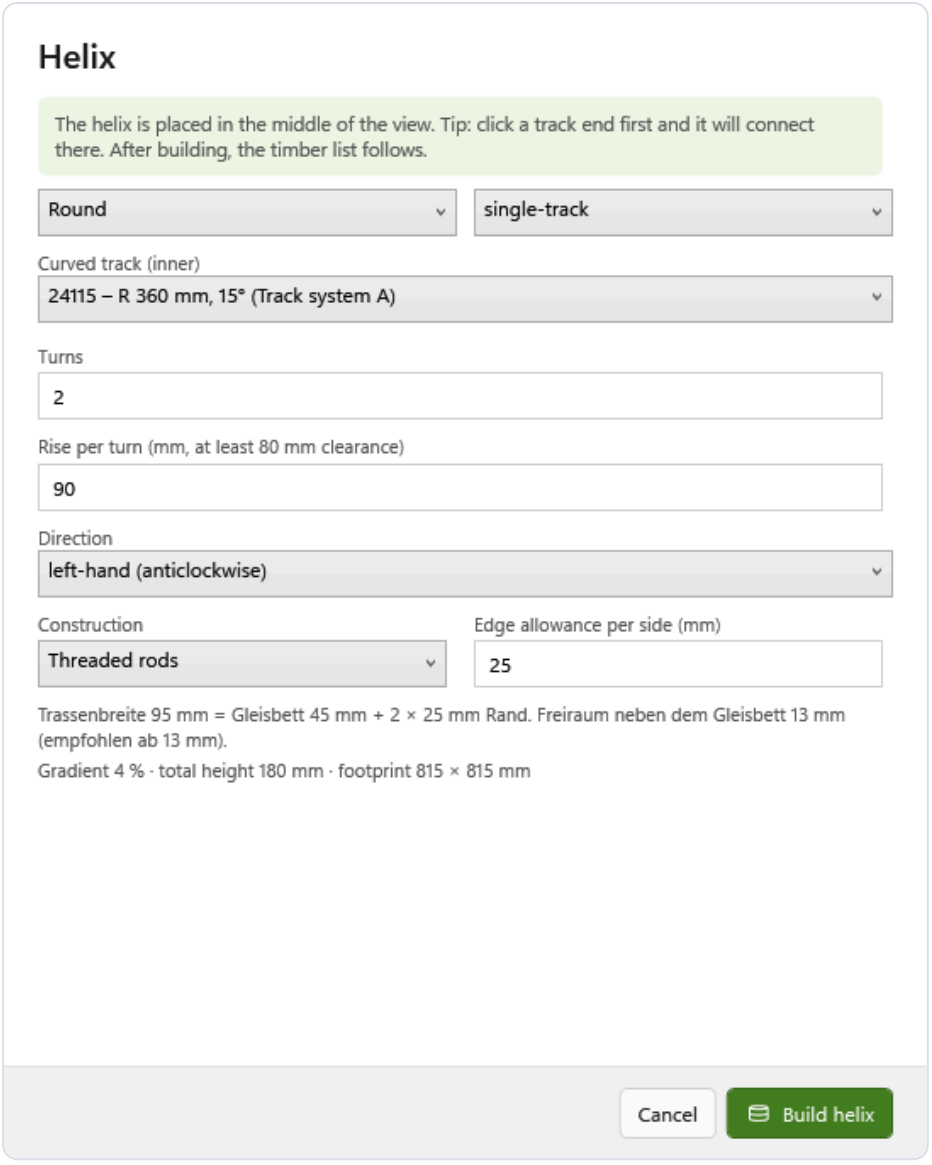


Figure 16 – Helix dialog with gradient, total height and space required.

1. Click a free track end if the helix should connect there. Without an active end it is created in the middle of the view.
2. Choose **Tracks › Helix…**.
3. Choose the **shape**: *Round* or *Oval* – with the oval, straight tracks lie along the two long sides; choose the straight track and the number per side.
4. Choose *single-track* or *double-track*. With double track the second track lies outside: as a catalogue curve with a matching radius or as flexible track with a freely chosen track spacing.
5. Choose the **curved track** for the inner track, the number of **turns**, the **height per turn** and the **direction**.
6. Under **Height and levels** enter the **start height** (rail top at the start of the helix) and choose the **level of the first turn**. The defaults are the height of the active track end or the base height of the active level.
7. With *Put each further turn on its own new level* (on by default) the helix starts on the chosen level, and after each full revolution (360°) the tracks go onto a new level “Helix – turn 2”, “… 3”

and so on, with the height at the start of that turn. This lets you show, hide and edit the turns separately later. There are at most 10 levels.

8. Below the fields, gradient, total height, space required, start and end height and the number of new levels are shown continuously. Click **Build helix**.

The height per turn must at least reach the clearance height of the gauge, otherwise no train fits under the next turn; the dialog points this out. Turns lying on top of each other are not connected to each other.

Under **Construction** you choose how the track bed is supported: on **threaded rods** or by **rib construction** with slotted ribs inside and outside, in whose slots the track bed boards lie. The timber list then contains the ribs with the heights of all slot bottoms instead of the threaded rods.

Helix timber list PRO

After building, TrackMaster Pro shows the **timber list** – everything you need for the helix substructure, with finished cutting dimensions. It can be copied or saved as a CSV file for a spreadsheet.

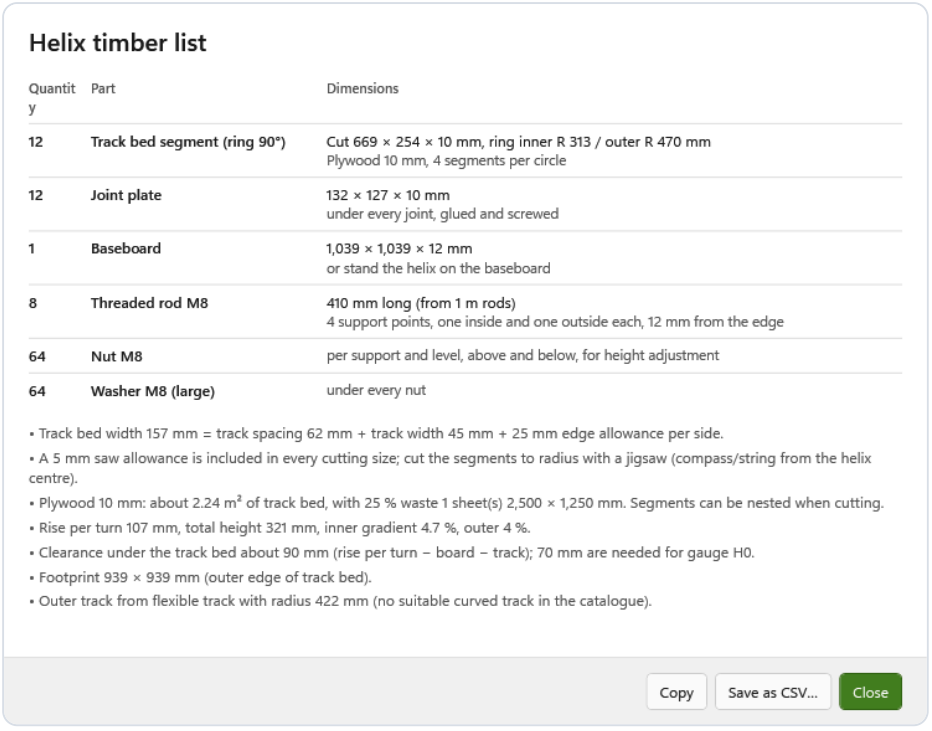


Figure 17 – Timber list of a double-track helix.

Part	Content
Track bed segments	ring segments of plywood with cutting dimensions (length × depth × thickness) and inner and outer radius. 4, 6, 8 or 12 segments per circle, so that none is longer than 1200 mm.
Straight track bed boards	with the oval, one board per long side and turn
Joint plates	under every joint, for gluing and screwing

Base plate	with a 50 mm margin all round – or stand the helix directly on the baseboard
Threaded rods, nuts, washers	one rod inside and outside per support point; nuts and large washers above and below each level, so that the height can be fine-tuned
Notes	track bed width, sheet requirement, gradient inside and outside, clear height and space required

Allowances

- **Track bed:** about 2.7 times the track gauge per track; with double track plus the track spacing.
- **Edge allowance:** at least 25 mm per side (for large gauges 1.5 times the track gauge) – room for the threaded rod hole 12 mm from the edge and for a guard rail.
- **Saw allowance:** 5 mm, already included in every cutting dimension.
- **Wood thickness:** 8 mm for small gauges, 10 mm up to gauge 0, 15 mm for large gauges; base plate at least 12 mm.
- **Threaded rods:** M8, M10 for large gauges. The length results from total height, base plate and projection.
- **Sheet requirement:** calculated in sheets of 2500 × 1250 mm with 25 % waste. The segments can be nested when cutting.

> **Tip**
> Mark out the segments with a string compass from the centre of the helix and cut them with a jigsaw. Drill the holes for the threaded rods with a template so that all levels lie exactly on top of each other.

> **Caution**
> If the clear height under the track bed is tight, the list shows a warning. Then increase the height per turn or choose thinner wood.

Cutting template as SVG

Cutting template as SVG in the timber list window saves each track bed segment, each straight board and each joint plate as a separate SVG file at full size (1:1), together in a ZIP file, with the track centre drawn as a dashed line. Radii, number of segments and the dimensions of boards and plates can be adjusted beforehand. The files are suitable for printing as a template, for a plotter or a CNC router.

Rib construction of the layout PRO

In open-grid benchwork the roadbeds lie on ribs that stand on the frame at fixed spacing. **Tracks › Rib construction…** calculates these ribs from the track heights of the plan.

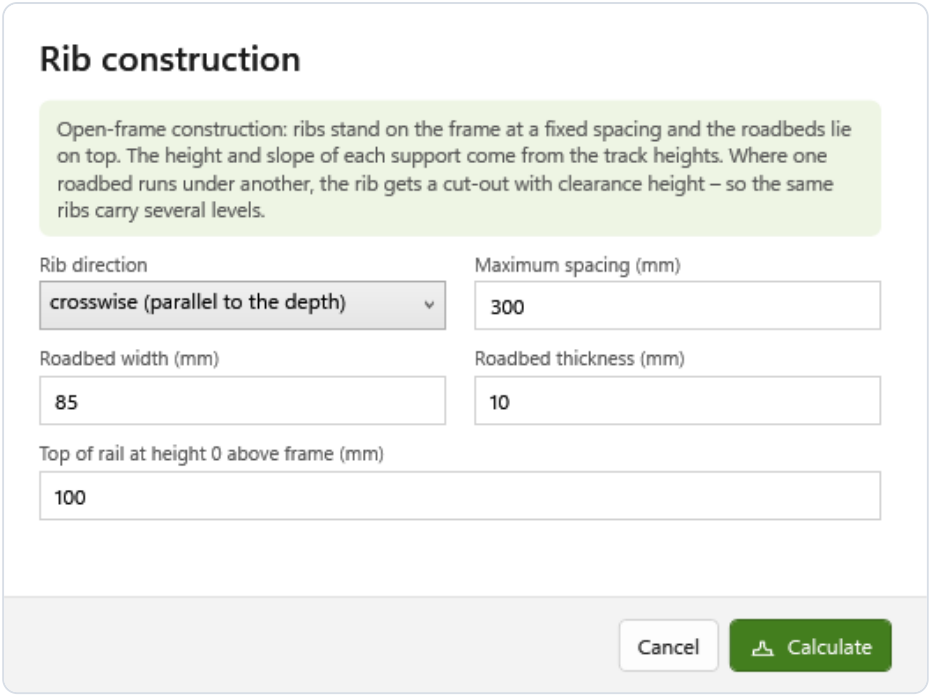


Figure 18 – Settings for rib construction.

- **Rib direction:** crosswise (parallel to the depth) or lengthwise (parallel to the width), with the maximum spacing in mm.
- **Roadbed width and thickness** as well as the **top of rail at height 0** above the frame.
- Each support gets its height and slope from the track heights; next to the supports the top edge falls away at 45°.
- If one roadbed lies under another, the rib gets a **cut-out** with the clearance height of the gauge – so the same ribs carry several levels.

The result shows each rib with position, length, height, supports and cut-outs. It can be copied, saved as CSV and output as SVG: an overview and each rib as a 1:1 template in its own file, together in a ZIP file.

Roadbed from selection PRO

For a ramp or a freely routed line, **Tracks › Roadbed…** calculates the roadbed board under the selected tracks:

1. Select the tracks, for example a gradient.
2. Choose **Tracks › Roadbed…** and enter roadbed width, thickness, maximum part length and joint plate length.
3. The roadbed follows the track centre and is divided into parts of equal length with joint plates. Save the parts as SVG: each part and each plate as its own file at full size, with the track joints and their heights drawn in.

Tunnel

Select tracks and click **Tracks › Tunnel**. Tunnel tracks are drawn below the other tracks in the plan, get portals in 3D and count as permitted under terrain in the plan check.

10 Scenery

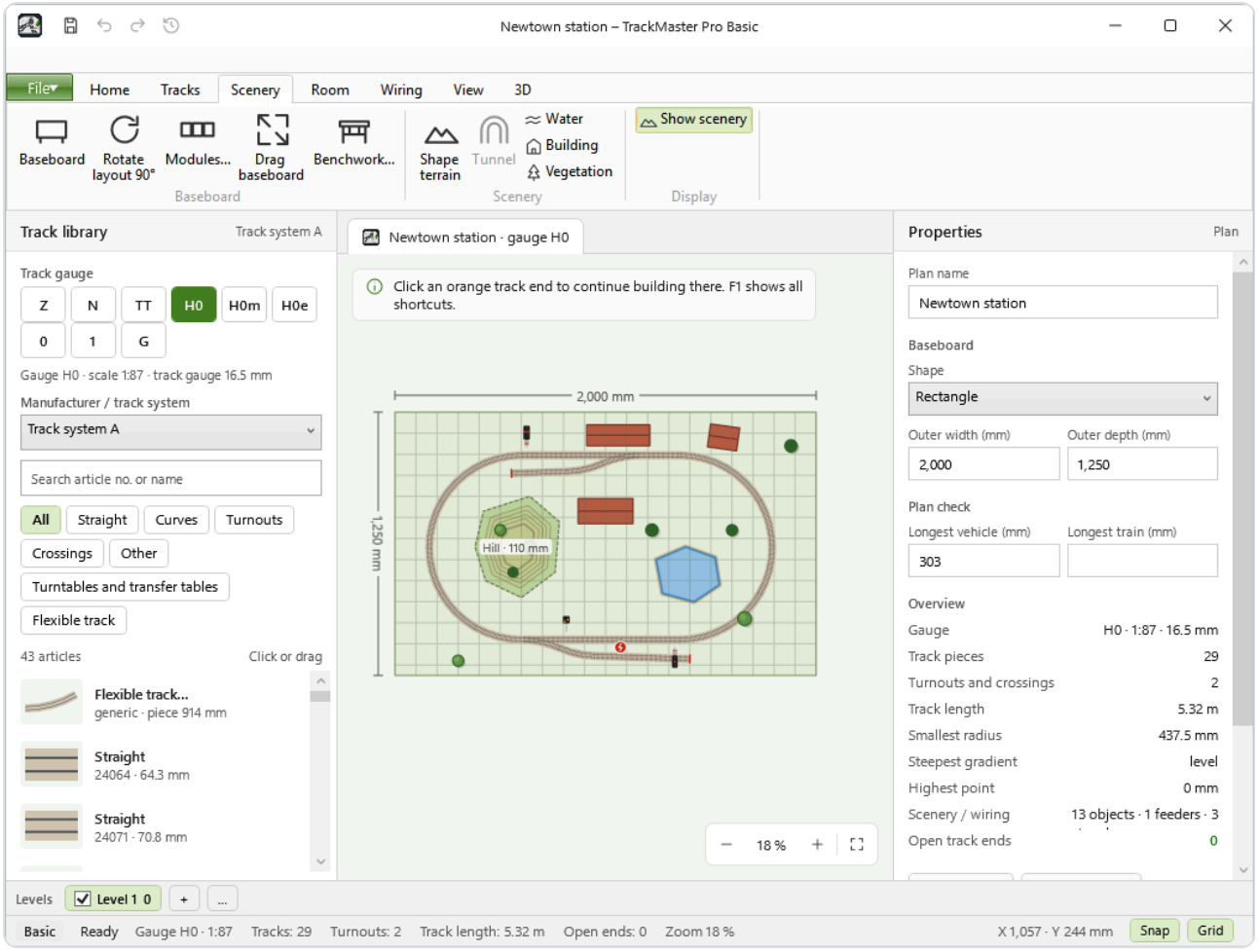


Figure 19 – Scenery tab with hill, lake, buildings and trees.

Tool	How it works
Shape terrain	Clicks set the corner points of a height area (hill, mountain, plateau). Enter closes the area, Esc cancels. Enter the height in the properties.
Water	Draw a lake or river as a polygon, as with terrain; the depth is in the properties.
Building	Draw a rectangle; width, depth, height and rotation can then be entered exactly.
Vegetation	Each click places a tree; diameter and height are adjustable.

Every object has a name and can be clicked, moved and deleted with Del. **Scenery › Show scenery** hides the scenery when you only want to see the tracks. The plan check reports tracks lying in buildings, in water or under terrain without a tunnel.

How the terrain is shown

- **Hills and mountains** rise from the edge of the area with a natural slope up to the height entered. In the track plan, height bands and contour lines show how steep the terrain is, as on a hiking map; every fifth line is bolder. The line spacing depends on the height (for example every 20 mm).
- **In the 3D view** gentle slopes are meadow, steeper ones earth and very steep ones rock. The meadow on the baseboard has several shades of green.
- **Water** has a shore strip. **Trees** are deciduous or conifers, shown with shadows in the track plan.
- Buildings, trees and piers stand on the terrain – a tree on the hill therefore stands on top of the slope.
- **Terrain and levels:** a hill stands on the base height of its level and grows with it. In the 3D view with scenery, raised tracks get an **embankment** sloping down to the ground; over other tracks piers remain, so that a train fits underneath.

11 Room and background image

Will the layout actually fit into the room? Can the door still open, and can you reach every spot? In the **Room** tab you draw your room with doors, windows and furniture around the layout. A background image – such as a floor plan or an old track plan – helps you trace.

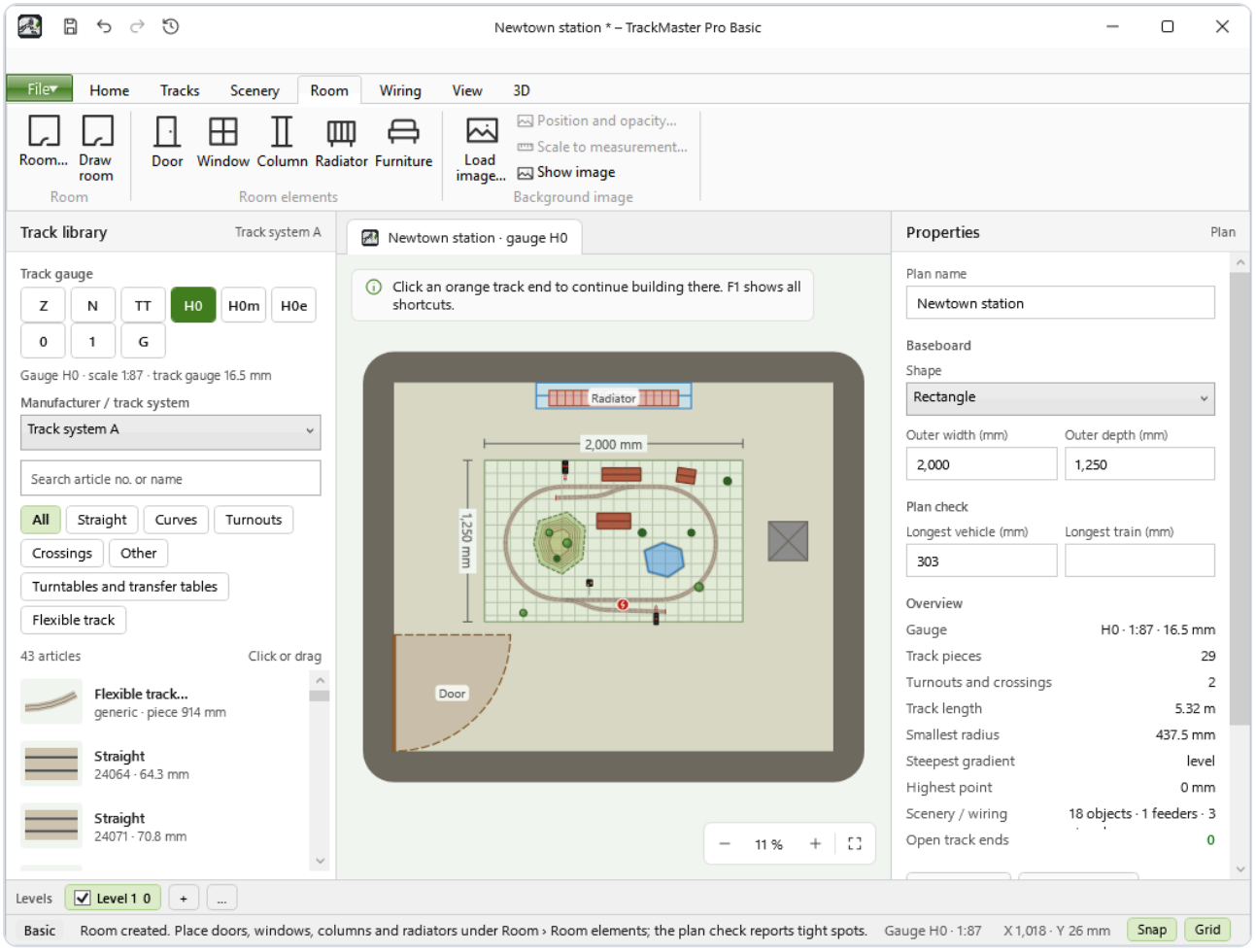


Figure 20 – Room tab: room with door, window, radiator and column around the layout.

Creating a room

For a rectangular room one dialog is enough:

1. Measure the room inside, from wall to wall, and note where the layout is to stand.
2. Click **Room › Room...**.
3. Enter the **inner width**, **inner depth** and **wall thickness** (all in mm).
4. Under **Inner corner X** and **Inner corner Y** state where the top left corner of the room lies – measured from the top left corner of the baseboard. If the board stands 50 cm from the left wall and 30 cm from the top wall, enter **−500** and **−300**. Negative means: to the left of or above the board.
5. Click **Create**. The walls appear in grey around the board.

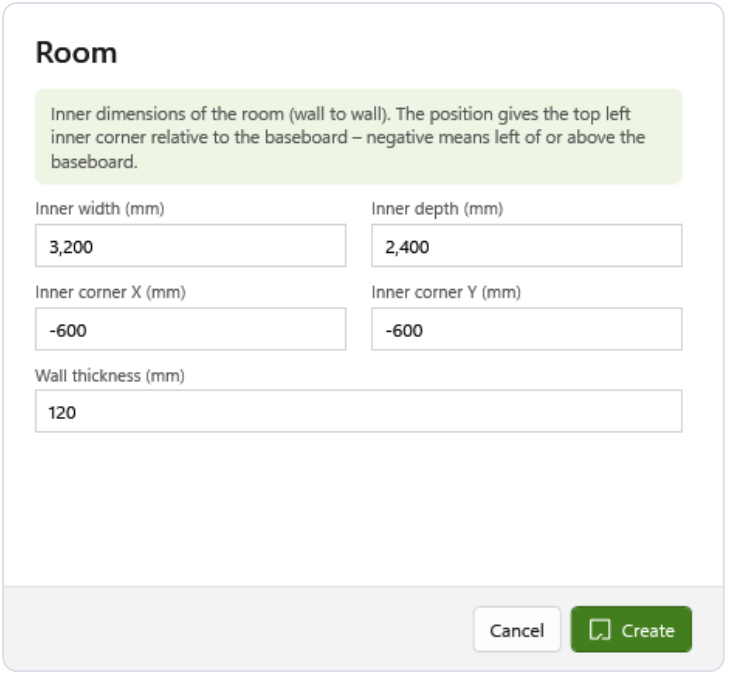


Figure 21 – Room dialog with inner dimensions, wall thickness and position.

If the room has alcoves or slanted walls, draw it with **Room › Draw room**: click all corners of the **inner face of the wall** one after the other. A double-click or Enter closes the room, Esc cancels. You select a room by clicking a wall; the properties show its name, wall thickness and floor area in m².

Doors, windows and furniture

The **Room › Room elements** group contains **Door**, **Window**, **Column**, **Radiator** and **Furniture**. Choose the element, click the spot in the plan or drag a rectangle of the size you want. Then enter width, depth, height and rotation exactly in the properties; the type can also be changed there afterwards. For a door the plan shows the **swing area** – the space the door leaf needs when opening. Esc ends the tool.

What the plan check reports in the room

Finding	Meaning
Board over the wall	**Error:** the baseboard projects beyond a wall – it will not fit into the room like this.
Door cannot open	**Error:** the swing area of a door lies under the board.
Room element under the board	**Warning:** a column, radiator or furniture stands under or against the board. Windows may lie under the board.
Aisle too narrow	**Note:** less than **60 cm** remains between board and wall. That much space is comfortable for building and operating. If the board stands right against the wall (up to 15 cm away), this is treated as intended and not reported.

> **Tip**
> Rooms and room elements are included in the DXF export (layer RAUM) – handy if you continue planning in a CAD program.

Loading a background image

An image under the plan helps you rebuild an existing track plan from a book or plan the layout to a floor plan.

1. Click **Room › Load image…** and choose a photo or scan (PNG, JPEG, GIF or WebP). Very large images are reduced when loading so that the plan stays manageable.
2. The image now lies semi-transparent under the plan. Under **Position and opacity…** you set where the top left corner lies (**Left**, **Top** in mm), how **wide** the image is and how strongly it appears (**Opacity** in %). There you can also replace it with another image or remove it.
3. **Show image** hides it and shows it again. The image is saved with the plan.

Scaling the image to a measurement

To give the image exactly the right scale, measure a distance on it whose real length you know – such as a wall of the room or a track:

1. Switch on **Home › Dimension** and click the start and end of the known distance in the image.
2. Click the new dimension line so that it is selected.
3. Click **Room › Scale to measurement…**. The dialog tells you how long the line currently is in the image. Enter the **real length** in mm and click **Scale**.

The image is enlarged or reduced to fit. You can then delete the dimension line and lay your tracks exactly on the image.

12 Wiring, signals and digital

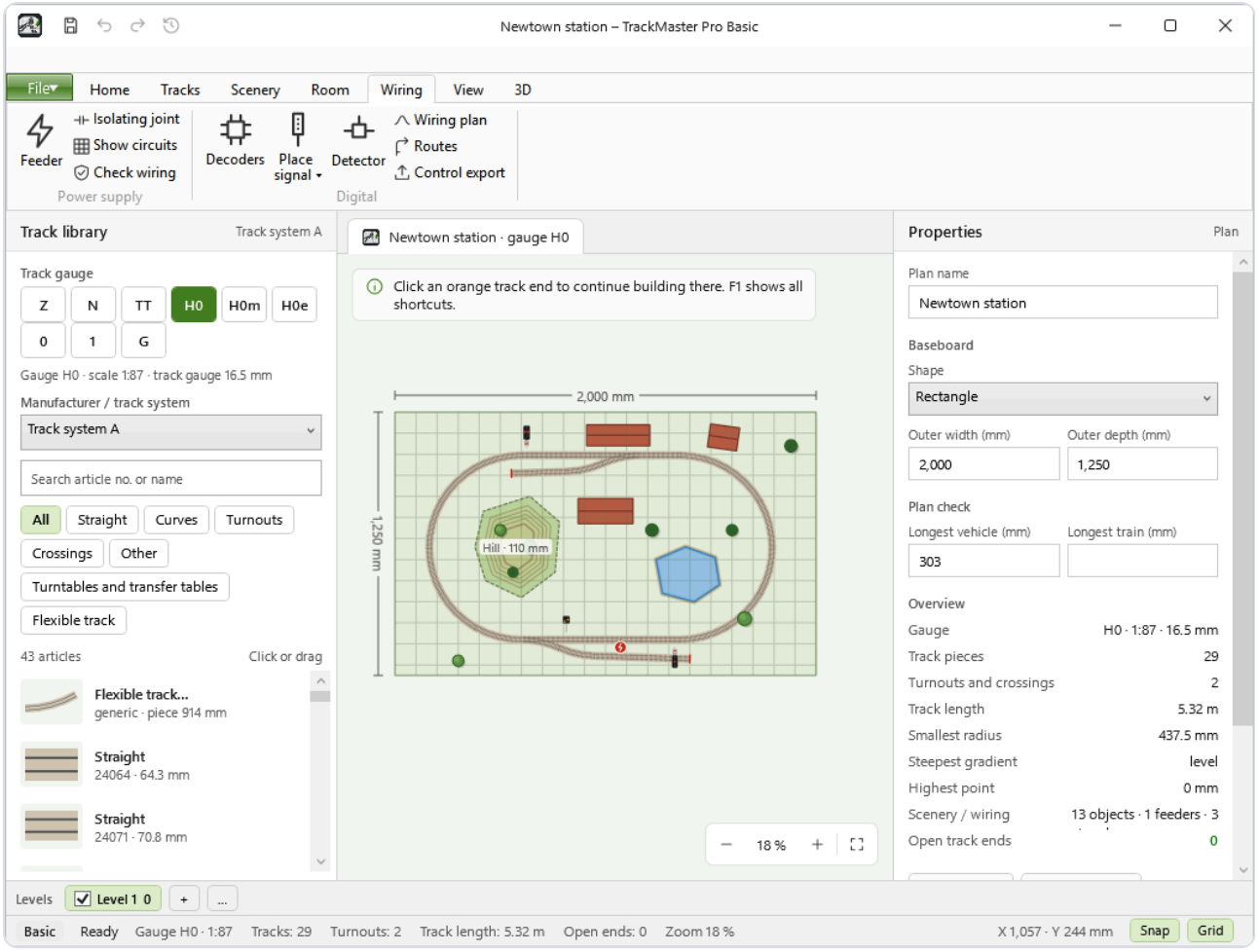


Figure 22 – Wiring tab with feeders, isolating joints and signals.

Feeders and isolating joints

- **Wiring › Feeder**: a click on a track places a connection. In the properties you assign it to a circuit or booster.
- **Wiring › Isolating joint**: a click on a track joint places insulated rail joiners.
- **Wiring › Show circuits** colours every power section in a colour of its own.
- A second click with the same tool on an existing feeder or signal removes it again; delete selected objects with Del.

Checking the wiring

Wiring › Check wiring shows sections without a feeder, reverse loops and addresses used twice. TrackMaster Pro detects **reverse loops** automatically in two-rail systems – there a reverse loop circuit with isolating joints is required. For centre-rail track systems this check is omitted, because reverse loops need no special circuit there.

Signals

Under **Wiring › Place signal** you choose the type of signal and then click a track end. The signals are shown as light signals of the German H/V signal system with a prototypical signal head – in the plan, in the 3D view and in the selection. Each shows its stop aspect; dark lamps can be seen in their real positions.

Signal	Stop aspect	Signal head
Main signal	Hp 0 – stop	green at the top, red in the middle, yellow at the bottom (green + yellow = Hp 2, proceed slowly); red and white mast plate
Exit signal	Hp 00 – stop	two red lamps side by side, plus green, yellow and two white lamps for Sh 1 (shunting permitted)
Block signal	Hp 0 – stop	green at the top, red at the bottom
Distant signal	Vr 0 – expect stop	two yellow lamps rising to the right, two green (Vr 1), yellow + green (Vr 2)
Shunting signal	Sh 0 – stop, no movement	dwarf signal: two red lamps horizontally, two white rising to the right (Sh 1)

Decoders and addresses

Wiring › Decoders lists all turnouts and signals with their digital addresses. Addresses can be entered individually or assigned automatically; conflicts are marked. Turnouts with two motors (three-way turnout, double slip) use two addresses.

Detectors

Detectors tell a digital control system where a train is at the moment. Choose **Wiring › Detector** and then the type:

- **Occupancy detector** – reports an occupied track from the current drawn by the vehicles.
- **Contact track** – reports when a train passes a particular spot.
- **Reed contact** – triggered by a magnet under the vehicle.

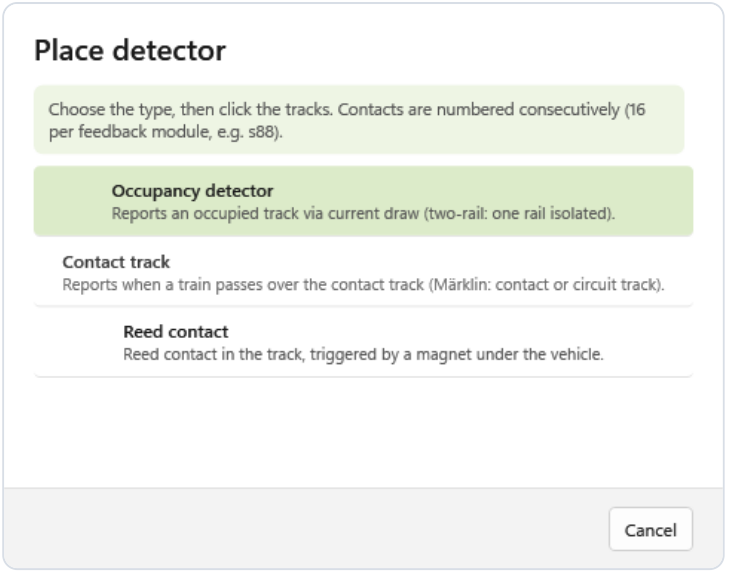


Figure 23 – Choosing the detector type.

Then click the tracks. The contacts are numbered consecutively – 16 per feedback module. Clicking an existing detector removes it. Select a detector to change its name, type and contact number in the properties. The detectors appear in the wiring plan with module and input, in the parts list together with the number of feedback modules needed, and in the export for control software.

Wiring plan

Wiring › Wiring plan summarises the wiring: power sections with their feeders, all isolating joints, the detectors and the decoder assignment. It can be saved as a text file in the Pro edition.

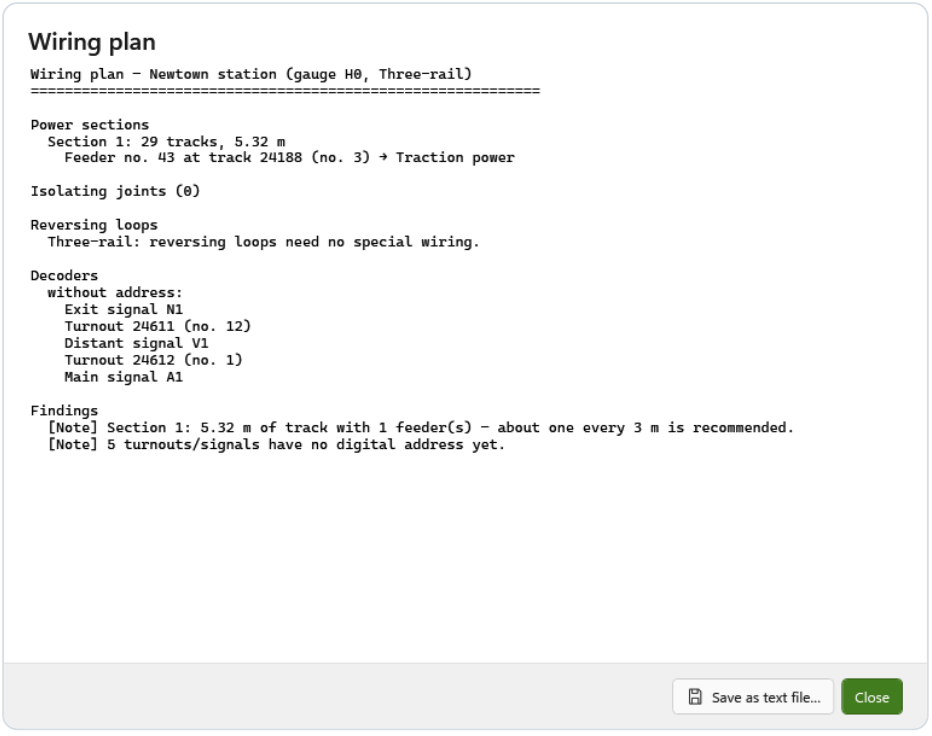


Figure 24 – Wiring plan.

Routes

Wiring › Routes finds all paths from one signal to the next signal in the direction of travel, to a buffer stop or to an open track end. Each route shows the turnouts with their position (straight, diverging, left or right), the length and the decoder addresses. Routes that need the same turnouts set differently are marked as a conflict.

Export for control software PRO

Wiring › Control export saves turnouts, signals, addresses, power sections and routes as a JSON file. From the routes window they can also be saved as text or CSV. This lets you take your planning into a layout control program without typing addresses twice.

13 Plan check

Home › Collision check (F7) examines the whole plan and lists all findings as error, warning or note. A click on an entry selects the tracks concerned and shows the spot.

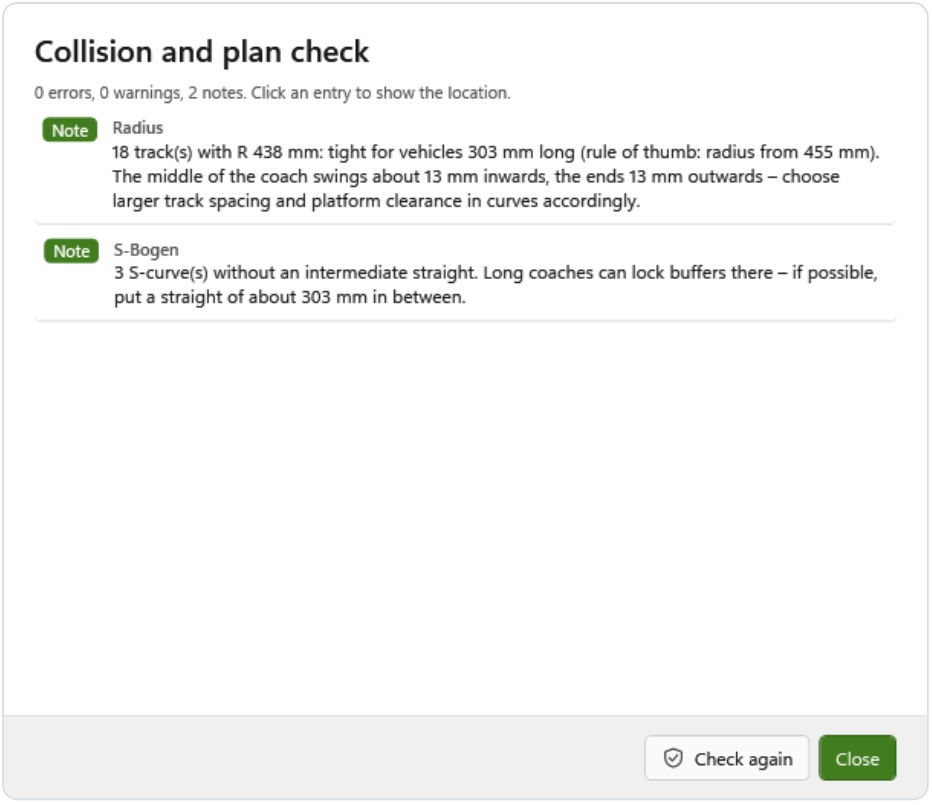


Figure 25 – Result of the plan check.

Check	What is reported
Track spacing and overlap	tracks that touch or lie too close together; for crossings at different heights the clearance height of the gauge is taken into account
Baseboard edge	tracks that project over the edge of the baseboard or into an open centre
Gradient	warning from 3 %, error from 5 %
Scenery	tracks in buildings, in water or under terrain without a tunnel
Open track ends	ends without a connection and without a buffer stop
Module joint	tracks that run across the joint between two modules of a modular layout
Radius for vehicles	curves that are too tight for the longest vehicle, with the overhang of the vehicle's centre and ends
Gradient in curve	warning if a curve climbs more steeply than 2 % – in a curve a train pulls noticeably harder

S-curve	reverse curves with a small radius and no straight in between, where long coaches can lock buffers
Room	board projects over a wall (error), door cannot open (error), column, radiator or furniture under the board (warning) – chapter 11
Aisle	less than 60 cm of space between board and wall
Usable length	sidings with a buffer stop that are shorter than the longest train

For the three vehicle checks, enter the **longest vehicle** in millimetres in the plan properties (section **Plan check**, field **Longest vehicle (mm)**). Without an entry the program assumes a long passenger coach at the scale of the gauge. Next to it is the field **Longest train (mm)**: enter the length of your longest train and TrackMaster Pro checks whether it fits on every siding. Left empty, this check is off.

> **Error, warning, note**
> An **error** must be fixed, otherwise the layout cannot be built or run like this. A **warning** should be checked. A **note** is advice – sometimes the spot is exactly as intended.

14 3D view

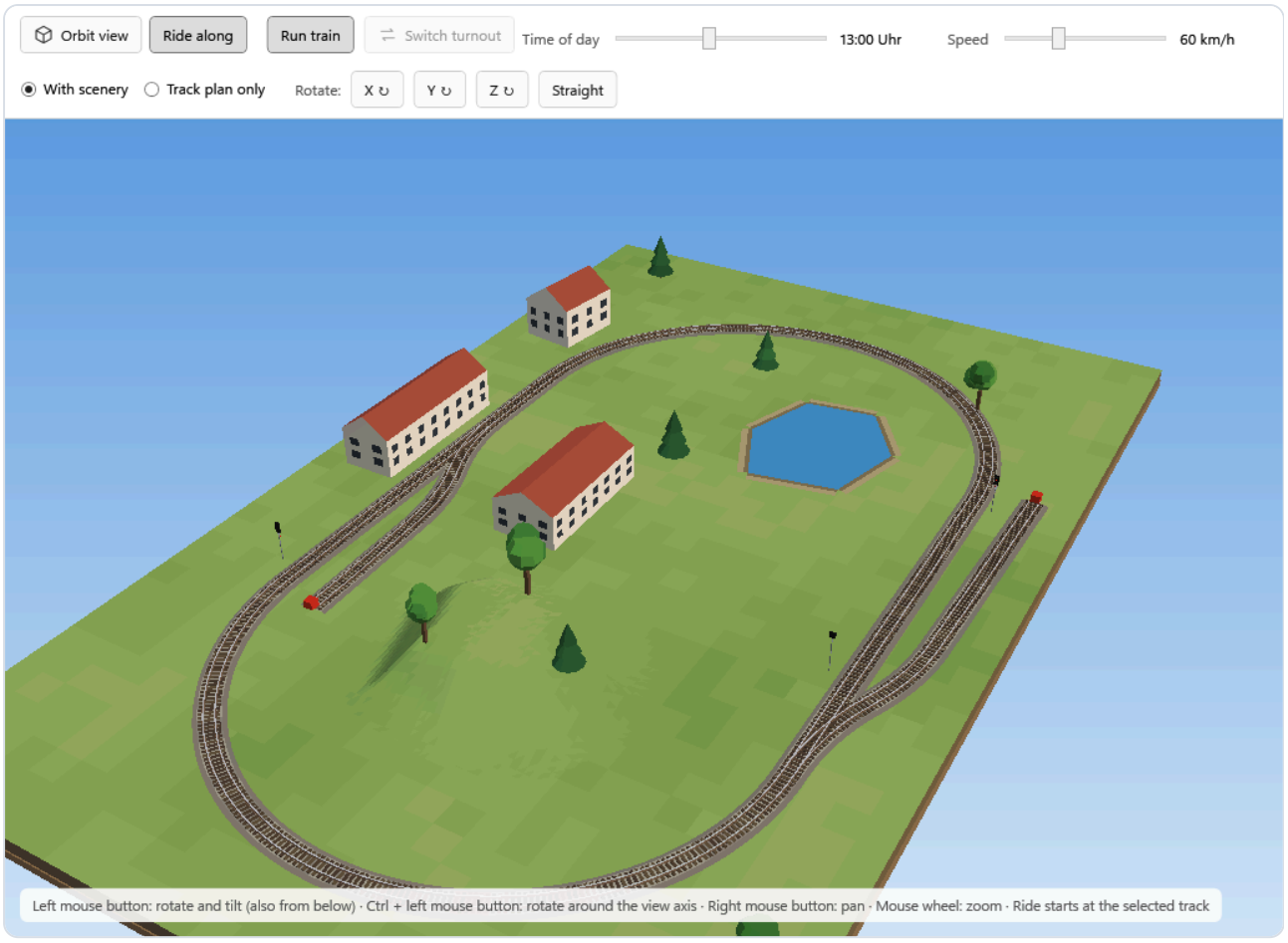


Figure 26 – The layout in 3D with terrain, buildings, trees and signals.

- **3D › 3D view**: dragging with the left mouse button rotates and tilts the layout – even below the baseboard –, Ctrl + left mouse button rotates around the viewing axis, the right mouse button pans, the mouse wheel zooms. The buttons **X**, **Y** and **Z** rotate in 90° steps around the transverse axis, the vertical axis and the viewing axis; **Normal position** resets the view.
- **With scenery** / **Track plan only**: show terrain, water, buildings and trees, or only baseboard, tracks and piers.
- **3D › Ride along**: you travel in the driver's cab over the tracks, starting at the selected track. You set the speed in prototype km/h. Before every turnout and double slip the possible **directions** appear at the top (◀ *left*, ▲ *straight ahead*, ▶ *right*); a click or the arrow keys ← ↑ → decide where the train goes at the next turnout. The status line names the turnout, its distance and the chosen direction. Only the route beyond the turnout changes, the train carries on without jumping. Plain crossings are always passed straight through.
- **3D › Time of day**: light from 5 am to 11 pm with the position of the sun, evening sky and lit windows.
- **Run train** in the 3D window: a train of a locomotive and three coaches runs over the layout. Here too the direction buttons or the arrow keys choose the route at the next turnout; W steps through the positions.

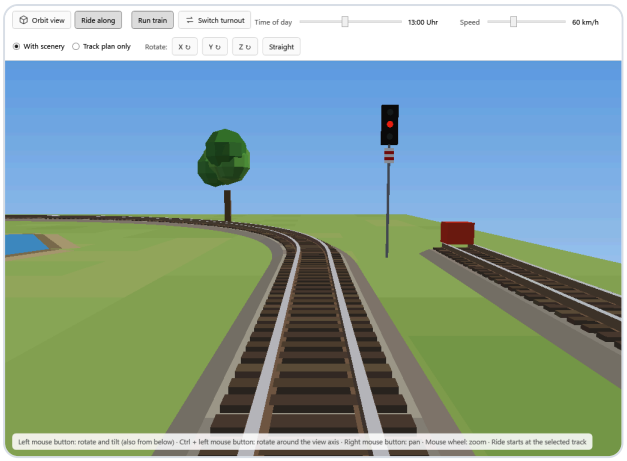


Figure 27 – Cab ride.

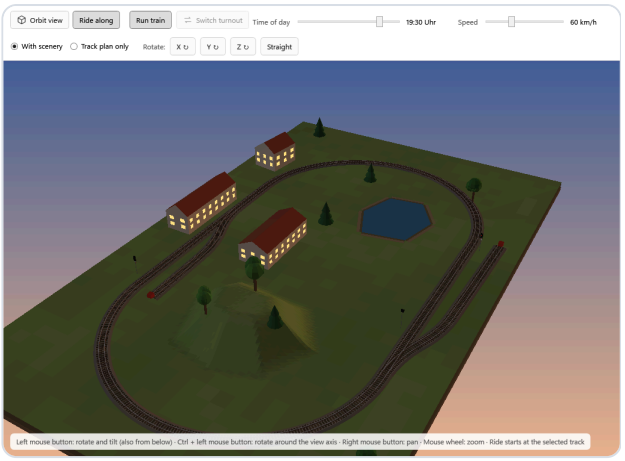


Figure 28 – Evening light.

The 3D view shows ballast bed, wooden sleepers and rails with a polished railhead, embankments and piers under raised tracks, tunnels, terrain, water, buildings, trees, signals and the benchwork. Hidden levels are missing here too. During the cab ride the view at cab height points slightly downwards, so that track, sleepers and turnouts ahead of the train are clearly visible.

15 Saving, parts list, printing and export

Saving small plans and opening work in both editions; the parts list, printing, the exports and saving large plans belong to the Pro edition.

Saving and opening

In the Basic edition, plans with up to 50 tracks can be saved; the Pro edition saves larger plans.

File › Save (Ctrl + S) and **Save as…** store the plan as a file with the extension **.trackmaster.json**. Opening (Ctrl + O) is also possible in the Basic edition. Plan files are interchangeable between Windows and Android and do not depend on the interface language. A plan file can also be opened with “Open with” or as a start parameter: `TrackMasterPro.exe MyLayout.trackmaster.json`.

Automatic backup: while you work, the program regularly backs up the plan in the background. If it was not closed properly – for example after a power cut – it offers to restore the backed-up state at the next start. The backup does not replace saving.

Parts list PRO

Home › Parts list counts all tracks by article number, with description and quantity, and adds the accessories: turnout motors, signals, feeders, insulated rail joiners, decoders, detectors and feedback modules. Flexible track is counted in whole lengths.

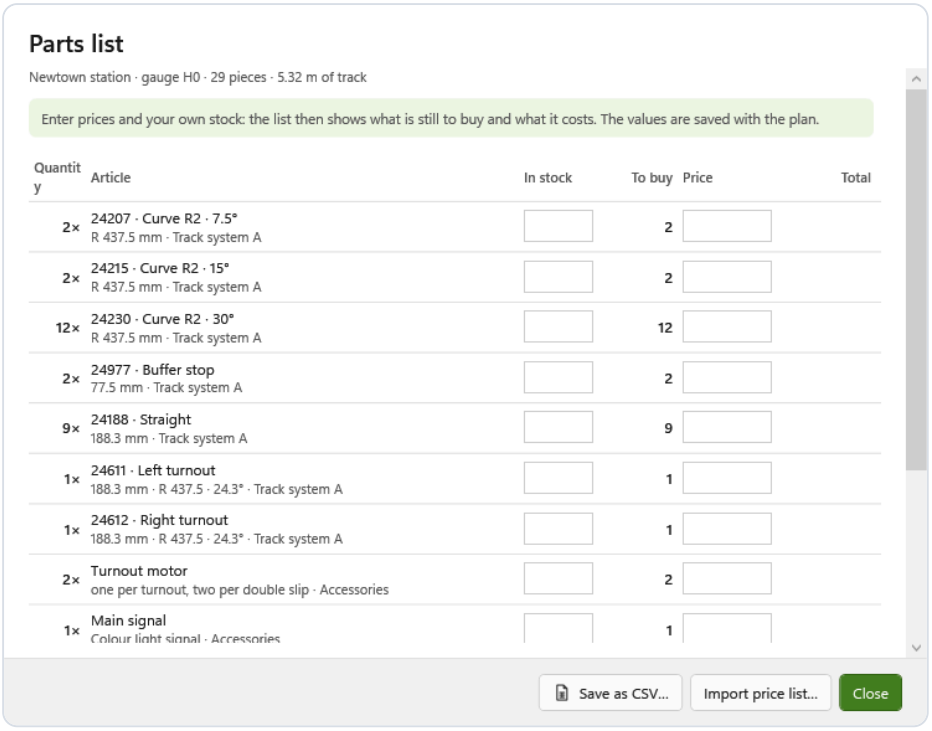


Figure 29 – Parts list with stock and prices.

What does the layout cost?

1. In the **In stock** column, enter how many pieces you already own. The **To buy** column shows what is still missing.

2. In the **Price** column, enter the unit price. Below, the costs are updated as you type: **Still to buy** (only what is missing), **All bought new** and how many articles are still **without price**.
3. Click **Close** – prices and stock are saved with the plan and are there again the next time you open it.

Importing a price list: instead of typing in all prices, use **Import price list…** to read a text or CSV file, for example from a spreadsheet. Each line contains the **article number** (or the description) at the front and the **price** after it, separated by semicolon, tab or comma:

```
10101; Straight 188 mm; 2.49
10202; Curve R2 30°; 2.79
Turnout motor; 19.90
```

Articles that do not occur in the plan are skipped. With **Save as CSV…** you take the whole list including stock and prices into a spreadsheet – as a shopping list for your dealer.

Printing PRO

File › Print… (Ctrl+P) offers two kinds:

- **Overview**: the plan with title, baseboard and summary, reduced to one page.
- **Building template 1:1**: the plan at full size on several pages. Neighbouring pages overlap by a glue strip; registration marks and the page labels (A1, A2, B1 …) help when gluing them together. Lay the template on the baseboard and transfer the track layout. Set the actual size (100 %) in the print dialog and measure the 100 mm test dimension on each page.

Via the Windows print dialog you can also print to a PDF file.

Export PRO

- **SVG at 1:1 scale** (**File › Export as SVG (1:1)…**): a vector graphic at full size, for example as a template for a plotter or for printing on several pages.
- **PNG image** (**File › Export as PNG image…**): the whole plan as a picture, about 4000 pixels wide – for showing by e-mail, in a forum or for printing.
- **DXF drawing** (**File › Export as DXF (CAD)…**): the plan at 1:1 in millimetres for CAD programs, CNC routers and laser cutters. The parts lie on separate layers: PLATTE (baseboard), GLEISACHSE (track centre line), SCHIENEN (rails), PRELLBOCK (buffer stops), GELAENDE (scenery), RAUM (room) and TEXT.
- **Parts list as CSV** (**File › Parts list as CSV**).
- **Timber list as CSV** from the timber list window.
- **Wiring plan as a text file** from the wiring plan window.
- **Cutting templates as SVG** for helixes, ribs and roadbeds: one file per part at full size, together in a ZIP file.
- **Ribs as CSV** and the **material and cutting list of the benchwork** as CSV (the benchwork belongs to the Basic edition).
- **Export for control software** (chapter 12).

16 The Android app

The app offers the same features as the Windows program, adapted to touch operation.

At the top is the ribbon with the tabs **Home, Tracks, Scenery, Wiring, View, 3D** and **File**, at the bottom the track library as a bar. Properties, plan check, helix, signals and the 3D view open as separate windows. In the Basic edition, Pro features carry a padlock. Choose design, font and font size under **View › Design and font** – with the same designs as on Windows. Copy and paste, routes, the language (**File › Sprache / Language**) as well as the train and turnouts in the 3D view are available too. Above the track library is the level bar; with **Scenery › Drag baseboard** you drag corners and modules with your finger, and tapping opens the dimensions.

All the newer features are there too: the **Room** tab with room elements and background image (you pick the image with the file picker), **Tracks › Parallel track** and **Height profile**, lock, group and align in the **Home** tab, detectors in the **Wiring** tab, the parts list with prices and stock, and export as PNG and DXF under **File**. You draw a room like terrain: tap the corners and finish with **Done**.

Gesture	Effect
Tap a piece in the library	attach at the active track end
Tap	select; make a free track end active
Drag	pan the view; on a selected piece: move the piece
Two fingers	zoom and pan
Rotate with two fingers	rotate the view freely; tapping the compass straightens it again
Long press	add a piece to the selection; on a free area: selection rectangle
Double tap	properties; on a free area: show the whole plan

The current plan is kept when you switch apps or lock the device. You save and open files via the Android file picker – in device storage or in cloud storage. This lets you start a plan on the tablet while on the move and continue on the PC.

17 Keyboard shortcuts and mouse commands

Key	Function
Ctrl + N	New plan
Ctrl + O	Open
Ctrl + S	Save (Basic up to 50 tracks)
Ctrl + P	Print PRO
Ctrl + Z / Ctrl + Y	Undo / Redo
Ctrl + A	Select all
Ctrl + D	Duplicate
Ctrl + C / Ctrl + V	Copy / Paste
Del	Delete
R / Shift + R	Rotate 15° right / left
F	Mirror
A	Switch connector
E	Next open track end
Enter / Space (short)	Attach the last track again
Arrow keys	Move 1 mm (Shift: 10 mm)
+ / − / 0	Zoom in / Zoom out / Whole layout
F7	Plan check
Enter / Esc	Close area / End tool
F1	Show keyboard shortcuts
W (3D window)	Switch the next turnout ahead of the train

Mouse	Function
Click on a track in the library	attach at the active track end
Drag a track from the library	place freely, snaps to nearby ends
Click on an orange end	select as the active track end
Drag a track	move (with snapping)
Shift +click	add to the selection
Drag on a free area	selection rectangle
Right / middle mouse button	pan the view
Space + drag	pan the view
Mouse wheel	zoom
Ctrl + mouse wheel	rotate the view (5°, with Shift 1°)
Click on the compass	straighten the view
Drag baseboard: drag corner / module	resize / move module
Drag baseboard: click	type in dimensions, rotate module or layout
Draw room: clicks, double-click	set corners, close the room
Room element: click or drag a rectangle	place door, window, column etc.
Click on a track of a group	select the whole group
3D: Ctrl + left mouse button	rotate around the viewing axis

18 Questions and solutions

A track does not snap on.

Check whether **Snap** is switched on in the status bar. Tracks only snap onto **free** ends; if the end is already connected, choose the next open end with E.

An oval does not quite close.

Small gaps occur when straights of different lengths are mixed. Close the gap with **Tracks › Close gap** or draw a flexible track.

A note about the Pro edition appears when saving.

The Basic edition saves plans with up to 50 tracks. The Pro edition saves larger plans. Install your licence under **File › Licence** (chapter 3).

The licence is reported as invalid.

The file must not be changed – not even by opening and saving it in a text editor. If the licence is bound to a computer, it applies only there; if an expiry date is entered, only until that day. Contact the publisher with your machine ID.

The licence is valid but “not activated on this device”.

The licence already runs on two other devices, or the device had no internet connection for 14 days. Release another device or connect to the internet and click **Activate now** in the licence dialog (section Online activation).

Windows reports “Windows protected your PC” during installation.

Click **More info** and **Run anyway**. The message appears for programs that Windows has not yet seen often.

Android does not install the app.

Allow the app you use to open the file (for example “Files” or the browser) to install unknown apps. If an older version with a different signature was already installed, uninstall it first.

Tracks have suddenly disappeared.

Their level is probably hidden. Tick it again in the level bar at the bottom or choose **Show all** in the Levels dialog. If **Show scenery** is off, hills, water, buildings and trees are missing as well.

A track cannot be moved, rotated or deleted.

It is locked. Select it and click **Home › Lock**, or untick **locked** in the properties.

One click suddenly selects many tracks.

The tracks are grouped. **Home › Ungroup** separates them again.

The background image does not match my dimensions.

Draw a line over a known distance in the image with **Home › Dimension**, select it and call **Room › Scale to measurement...** (chapter 11).

The plan check says the door cannot open.

The swing area of the door lies under the board. Move the door, rotate it in the properties or make the board smaller at that spot.

How much does my layout cost?

Open the parts list (Pro), enter stock and prices or import a price list. The total is shown at the bottom of the list (chapter 15).

I have changed too much.

Home › History... shows the last 20 steps; a click jumps to an earlier state, and forward again.

The 3D view stays black or stutters.

Update the graphics driver. For very large layouts it helps to place trees and buildings sparingly at first.

The interface appears in the wrong language.

Choose **File › Sprache / Language...** and select the language. The menu item is labelled in both languages so it can always be found.

Where is my data?

Plans are where you saved them. Licence, settings and the automatic backup are in *%APPDATA%\TrackMaster Pro*. The program sends no data to third parties; at start-up it only fetches the current version number from the download page.

19 Glossary

Term	Meaning
Active track end	The free end marked in orange, where the next track is attached.
Apron, cross member	Outer frame and cross bracing of the benchwork on which the top board or the ribs rest.
Background image	Image under the plan, such as a floor plan or a track plan from a book, for tracing.
Baseboard	Surface on which the layout is built, usually a wooden board on benchwork.
Buffer stop	End piece of a track where the train stops.
Clear height	Free space between the top of rail and the underside of the track bed above.
Clearance height	Minimum height between two tracks lying on top of each other so that vehicles fit underneath; depends on the gauge.
Connector	One end of a track piece; a turnout has three, a crossing four connectors.
CSV file	Simple table file that any spreadsheet can open.
Curve	Curved track piece with a fixed radius and angle, for example 30°. Twelve 30° curves make a circle.
Detector	Tells the digital control whether a track is occupied or a train passes a spot.
DXF file	Drawing format for CAD programs, CNC routers and laser cutters.
Edge allowance	Addition to the track bed width for fixing and a guard rail.
Feeder	Power connection of a track section to a transformer, command station or booster.
Flexible track	Track supplied in lengths that can be bent to any radius and cut to length.
Gauge	Size of the model railway, for example H0, N or Z; sets track width and scale.
Gradient	How steeply a track climbs, in per cent: 2 % means 2 cm of height over 1 m of track.
Group	Tracks that are combined and are selected and moved together with one click.
Helix	Spiral climbing line for overcoming large differences in height on a small area.

Isolating joint	Insulated rail joint that separates two power sections.
Layout suggestion	Small layout the program builds from the quantities of track you entered (web version).
Level	Group of tracks and terrain objects with a common base height that can be shown and hidden together.
Module	Rectangular part of a layout with its own dimensions that can be combined with others into a layout.
Open track end	End of a track piece without a connection; marked orange in the plan. The plan check reports open ends without a buffer stop.
Oval	Simplest layout shape: two semicircles joined by straights. The train runs round endlessly.
Parallel track	Second track at a constant spacing next to a line.
Properties	Area on the right of the window; shows and changes the values of the selected track or object.
Radius	Distance from the centre of the circle to the middle of the track of a curve; the smaller, the tighter.
Reverse loop	Track arrangement in which a train returns to the line in the opposite direction; in two-rail systems this creates a short circuit that needs a special circuit.
Rib	Upright board on the frame on which the roadbed lies in open-grid benchwork.
Ribbon	Bar at the top of the window with all commands, arranged in tabs.
Roadbed	Board under the tracks that carries them at the desired height.
Route	Path from one signal to the next with the turnout positions required.
Saw allowance	Addition to the cutting dimension for the saw cut and finishing.
Scale	Ratio between model and prototype; in H0, 1 cm on the model is 87 cm in reality (1:87).
Siding	Track that ends at a buffer stop, where trains or wagons are parked.
Stop aspect	Signal indication that means “stop” (Hp 0, Hp 00, Vr 0, Sh 0).
Tab	One of the headings in the ribbon (Home, Tracks, Scenery, Room …); a click shows its commands.
Top of rail	Reference height of all height values in the plan.
Track system	Track range from one manufacturer; its pieces fit together exactly.
Turnout	Track piece where a track divides into two (or three) directions.

Usable length	Length of a track section between turnouts or up to the buffer stop – the longest train that can stand there.

TrackMaster Pro 1.5.0 · User manual, as of November 2026. Subject to change. All dimensions without guarantee; check the dimensions against the real product before cutting and buying.